

Important: Text generated by AI and may contain incorrect information, always check the content.

```
```python
Import necessary libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

Display the first few rows of the dataframe
display('First few rows of the dataframe:')
display(df.head())

Check the shape of the dataframe
display('Shape of the dataframe:')
display(df.shape)

Check for missing values
display('Missing values count:')
display(df.isnull().sum())

Since there are no missing values, we can proceed with summary statistics
display('Summary statistics for numerical columns:')
display(df.describe())

Check the distribution of the 'fetal_health' column
display('Distribution of fetal_health:')
display(df['fetal_health'].value_counts())

Plot histograms for all numerical columns
display('Histograms for numerical columns:')
for col in df.columns:
 if df[col].dtype != 'object':
 plt.figure(figsize=(10,6))
 sns.histplot(df[col], kde=True)
 plt.title(f'Histogram of {col}')
 plt.show()

Plot boxplots for all numerical columns
display('Boxplots for numerical columns:')
for col in df.columns:
 if df[col].dtype != 'object':
 plt.figure(figsize=(10,6))
 sns.boxplot(df[col])
 plt.title(f'Boxplot of {col}')
 plt.show()

Plot correlation matrix
display('Correlation matrix:')
plt.figure(figsize=(20,16))
sns.heatmap(df.corr(), annot=True, cmap='coolwarm', square=True)
plt.title('Correlation Matrix')
plt.show()

Check for highly correlated features
display('Highly correlated features:')
corr_matrix = df.corr()
upper = corr_matrix.where(np.triu(np.ones(corr_matrix.shape), k=1).astype(bool))
high_corr_features = [column for column in upper.columns if any(abs(upper[column]) > 0.7)]
display(high_corr_features)
```

'First few rows of the dataframe:'
```



| | baseline
value | accelerations | fetal_movement | uterine_contractions | light_decelerations | severe_decelerations | prolongued_decelerations | abnormal_short_term_variability | mean_value_of_short_term_variability | percentage_of |
|---|-------------------|---------------|----------------|----------------------|---------------------|----------------------|--------------------------|---------------------------------|--------------------------------------|---------------|
| 0 | 120.0 | 0.000 | 0.0 | 0.000 | 0.000 | 0.0 | 0.0 | 73.0 | 0.5 | |
| 1 | 132.0 | 0.006 | 0.0 | 0.006 | 0.003 | 0.0 | 0.0 | 17.0 | 2.1 | |
| 2 | 133.0 | 0.003 | 0.0 | 0.008 | 0.003 | 0.0 | 0.0 | 16.0 | 2.1 | |
| 3 | 134.0 | 0.003 | 0.0 | 0.008 | 0.003 | 0.0 | 0.0 | 16.0 | 2.4 | |
| 4 | 132.0 | 0.007 | 0.0 | 0.008 | 0.000 | 0.0 | 0.0 | 16.0 | 2.4 | |

5 rows × 22 columns

'Shape of the dataframe:'
(2126, 22)
'Missing values count:'

| | |
|--|---|
| | 0 |
| baseline value | 0 |
| accelerations | 0 |
| fetal_movement | 0 |
| uterine_contractions | 0 |
| light_decelerations | 0 |
| severe_decelerations | 0 |
| prolongued_decelerations | 0 |
| abnormal_short_term_variability | 0 |
| mean_value_of_short_term_variability | 0 |
| percentage_of_time_with_abnormal_long_term_variability | 0 |
| mean_value_of_long_term_variability | 0 |
| histogram_width | 0 |
| histogram_min | 0 |
| histogram_max | 0 |
| histogram_number_of_peaks | 0 |
| histogram_number_of_zeroes | 0 |
| histogram_mode | 0 |
| histogram_mean | 0 |
| histogram_median | 0 |
| histogram_variance | 0 |
| histogram_tendency | 0 |
| fetal_health | 0 |





dtype: int64
'Summary statistics for numerical columns:'

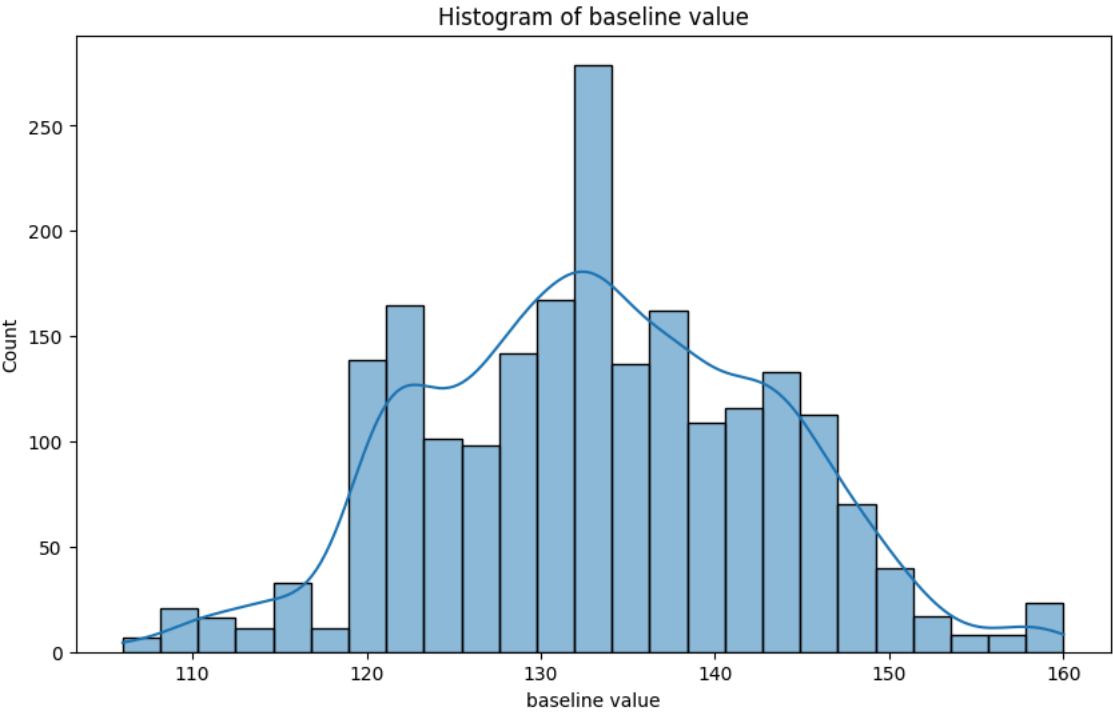
| | baseline
value | accelerations | fetal_movement | uterine_contractions | light_decelerations | severe_decelerations | prolongued_decelerations | abnormal_short_term_variability | mean_value_of_short_term_variability | pe |
|-------|-------------------|---------------|----------------|----------------------|---------------------|----------------------|--------------------------|---------------------------------|--------------------------------------|-------------|
| count | 2126.000000 | 2126.000000 | 2126.000000 | 2126.000000 | 2126.000000 | 2126.000000 | 2126.000000 | 2126.000000 | 2126.000000 | 2126.000000 |
| mean | 133.303857 | 0.003178 | 0.009481 | 0.004366 | 0.001889 | 0.000003 | 0.000159 | 46.990122 | 1.332785 | |
| std | 9.840844 | 0.003866 | 0.046666 | 0.002946 | 0.002960 | 0.000057 | 0.000590 | 17.192814 | 0.883241 | |
| min | 106.000000 | 0.000000 | 0.000000 | 0.000000 | 0.000000 | 0.000000 | 0.000000 | 12.000000 | 0.200000 | |
| 25% | 126.000000 | 0.000000 | 0.000000 | 0.002000 | 0.000000 | 0.000000 | 0.000000 | 32.000000 | 0.700000 | |
| 50% | 133.000000 | 0.002000 | 0.000000 | 0.004000 | 0.000000 | 0.000000 | 0.000000 | 49.000000 | 1.200000 | |
| 75% | 140.000000 | 0.006000 | 0.003000 | 0.007000 | 0.003000 | 0.000000 | 0.000000 | 61.000000 | 1.700000 | |
| max | 160.000000 | 0.019000 | 0.481000 | 0.015000 | 0.015000 | 0.001000 | 0.005000 | 87.000000 | 7.000000 | |

8 rows × 22 columns
'Distribution of fetal_health:'

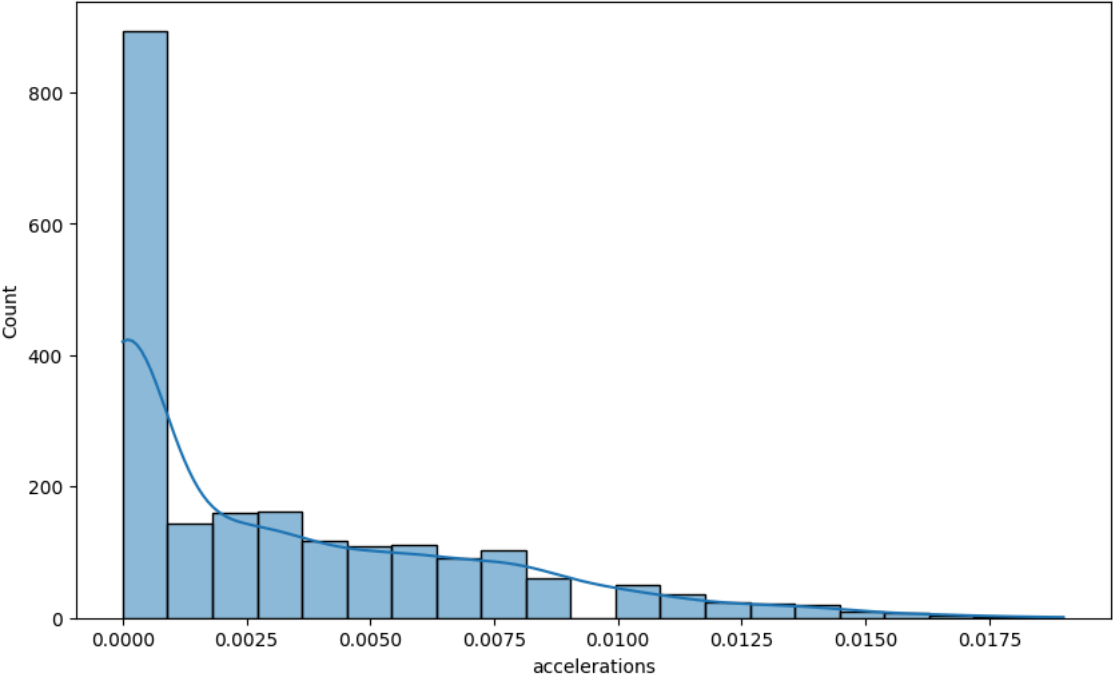
| | fetal_health |
|-----|--------------|
| 1.0 | 1655 |
| 2.0 | 295 |
| 3.0 | 176 |

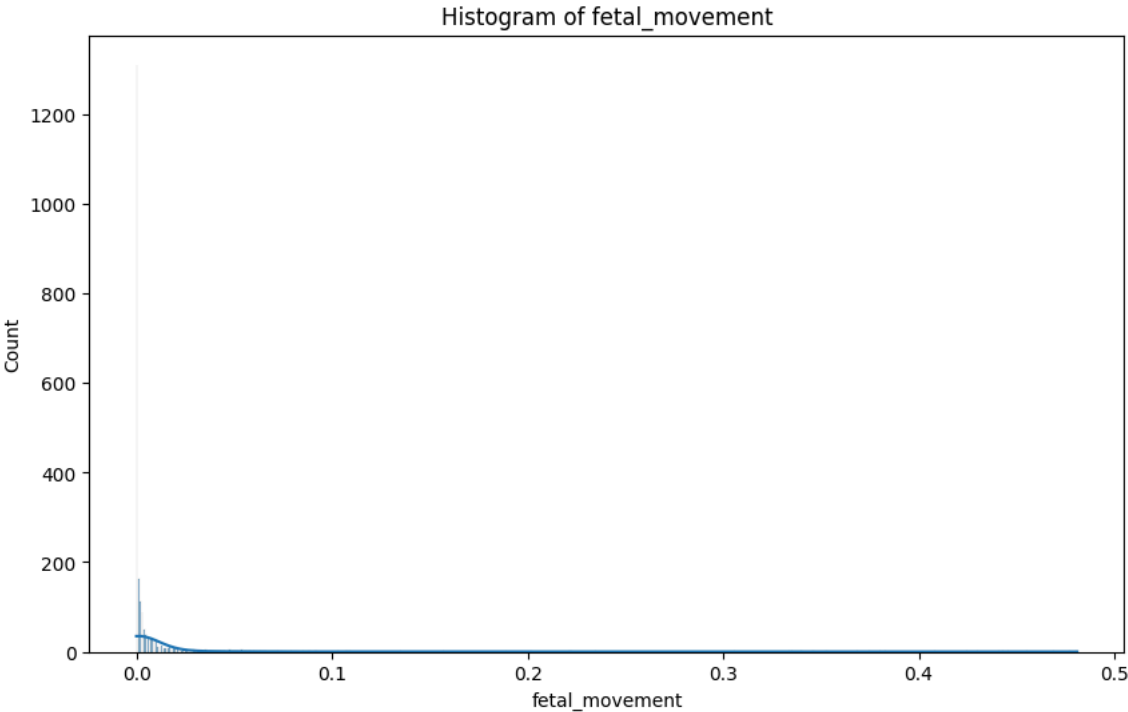
dtype: int64
'Histograms for numerical columns:'

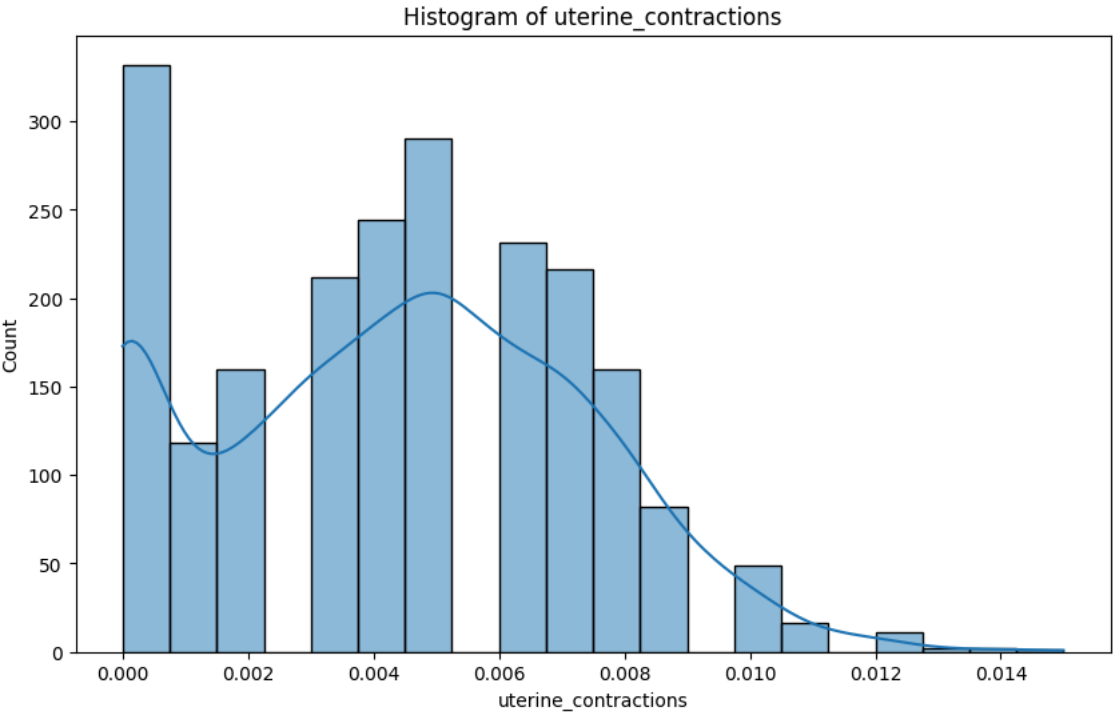


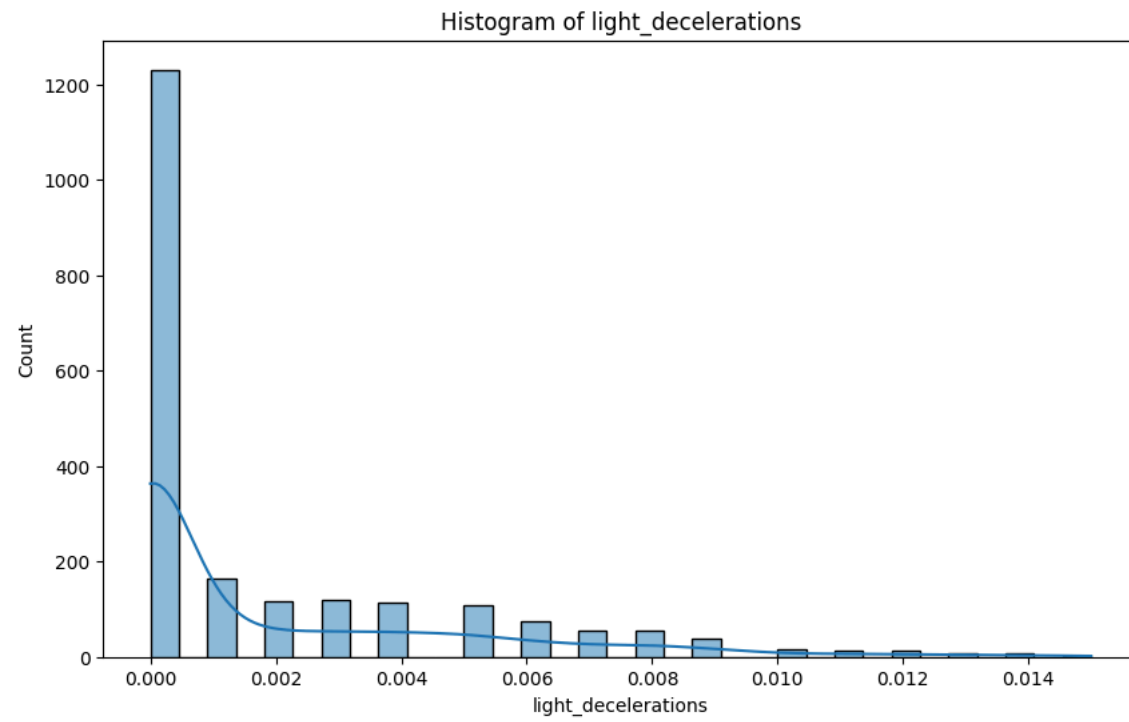


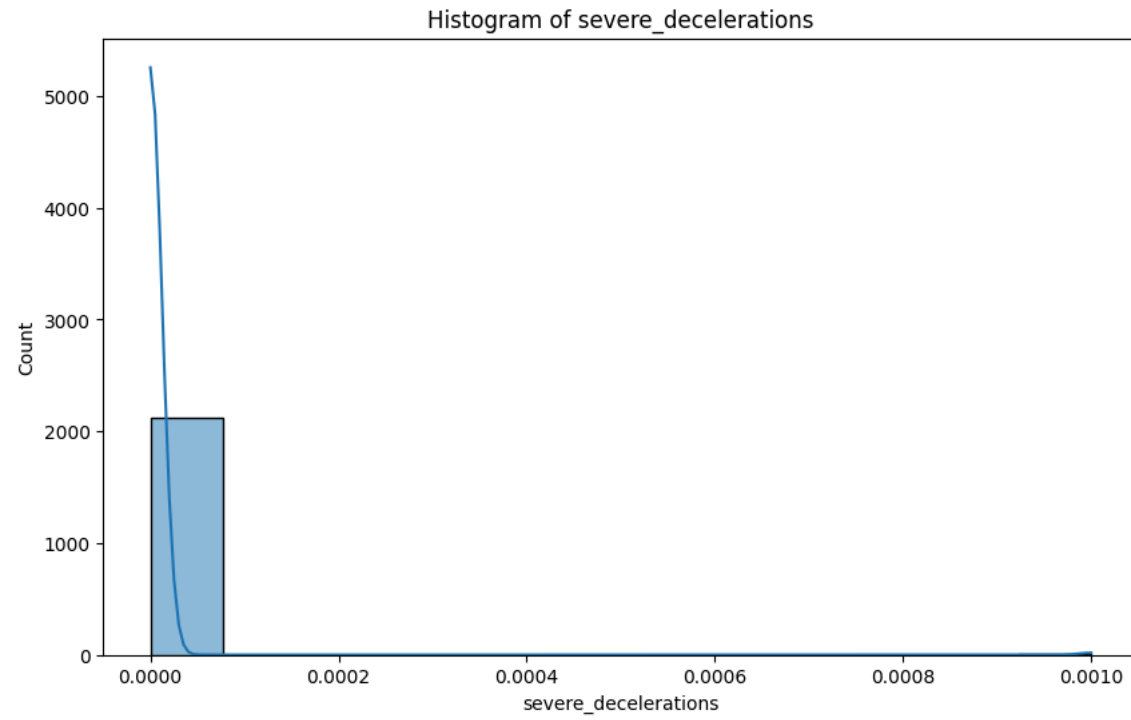
Histogram of accelerations

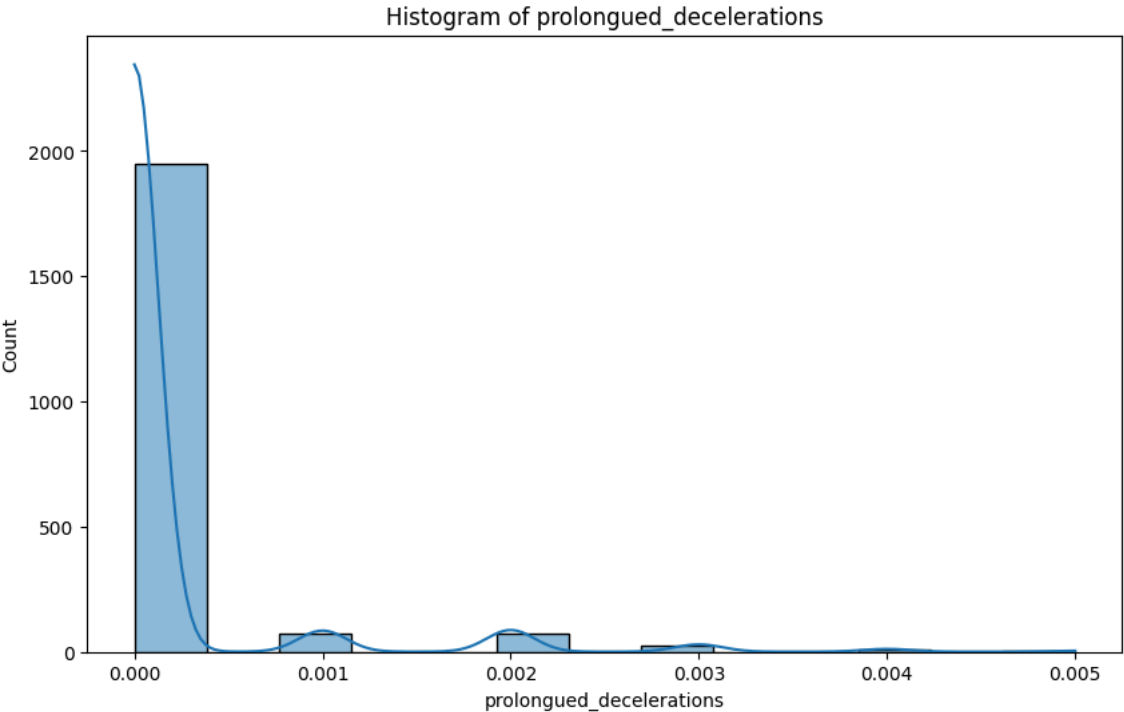


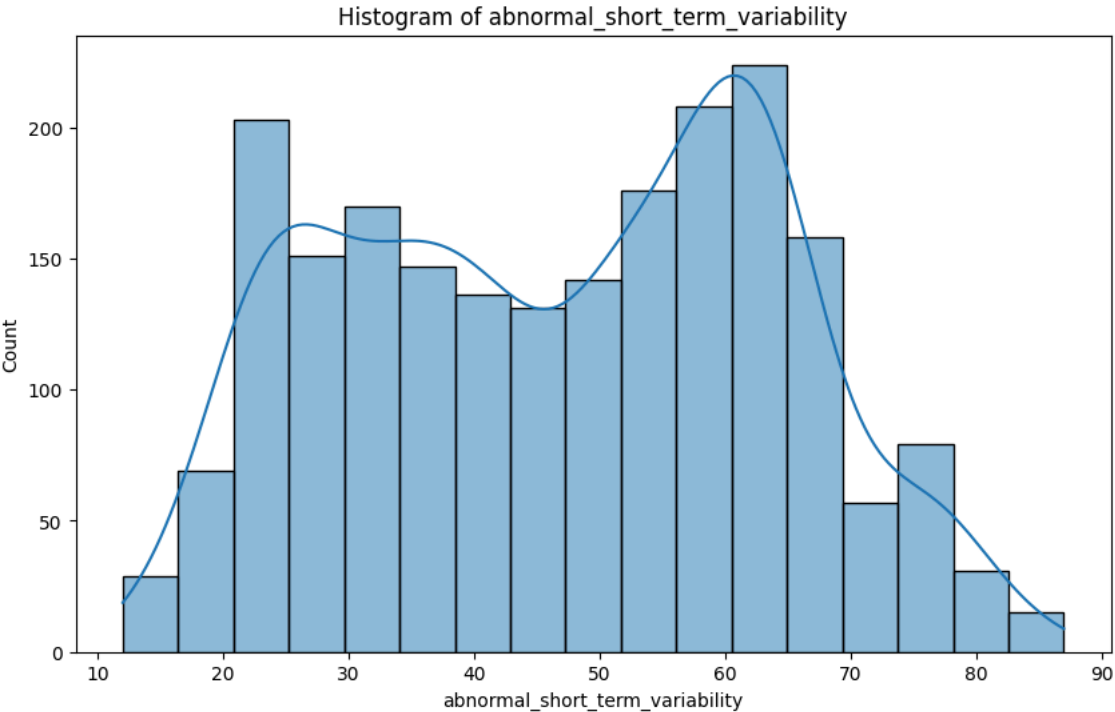


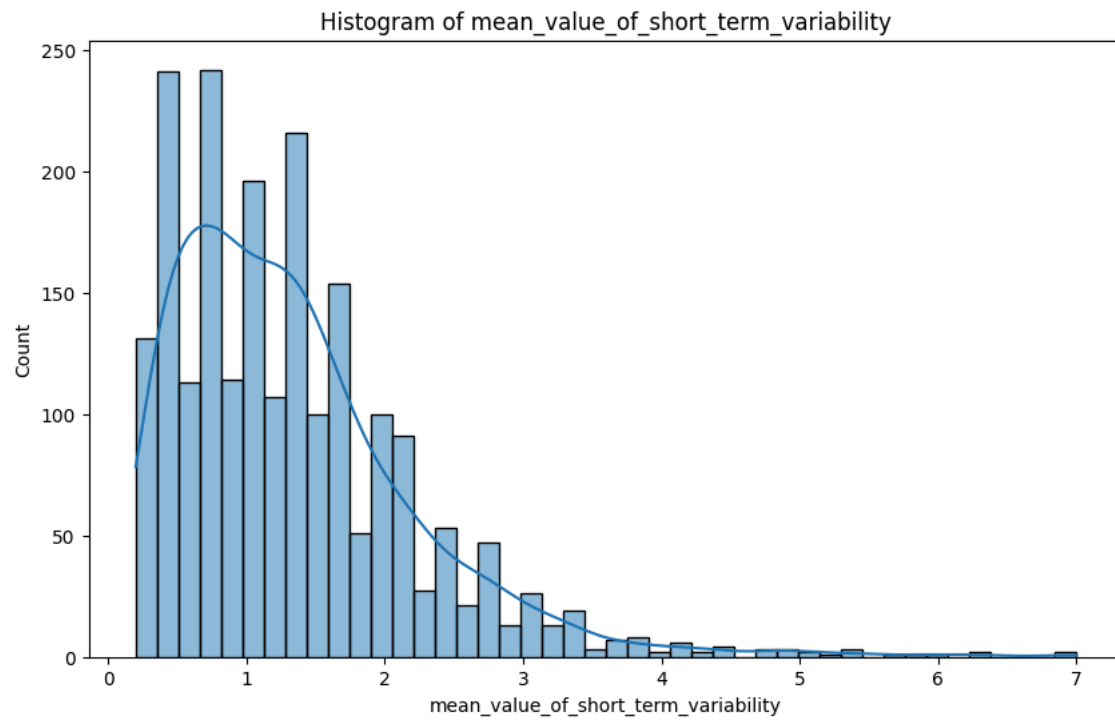


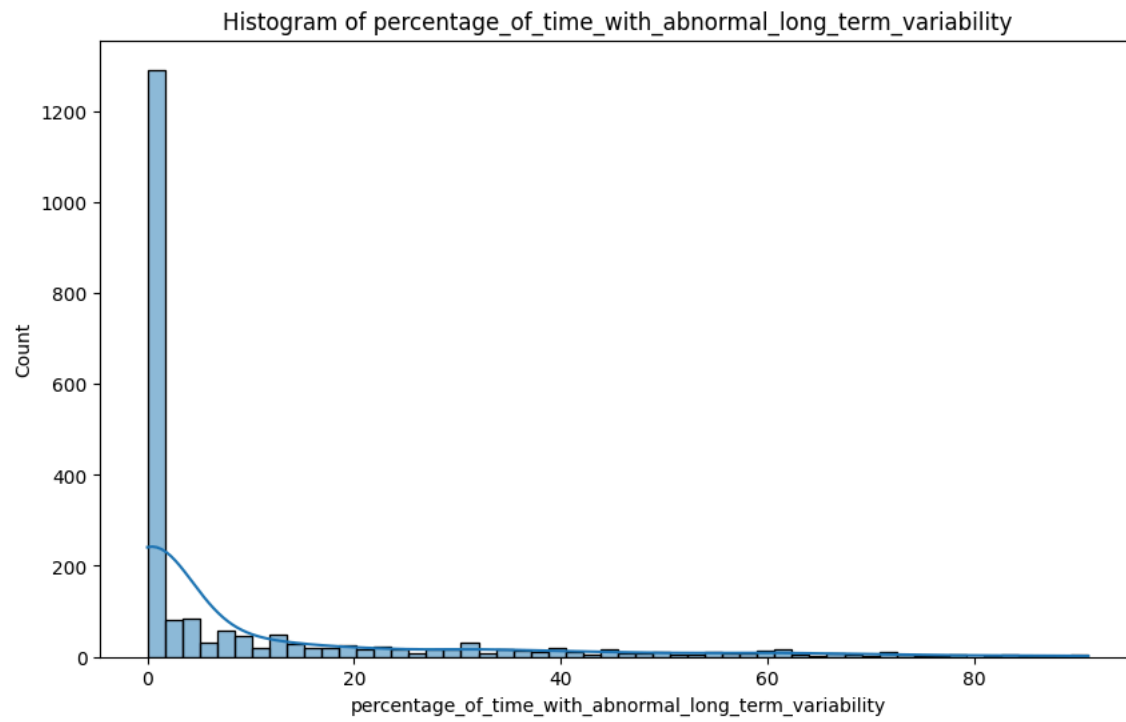


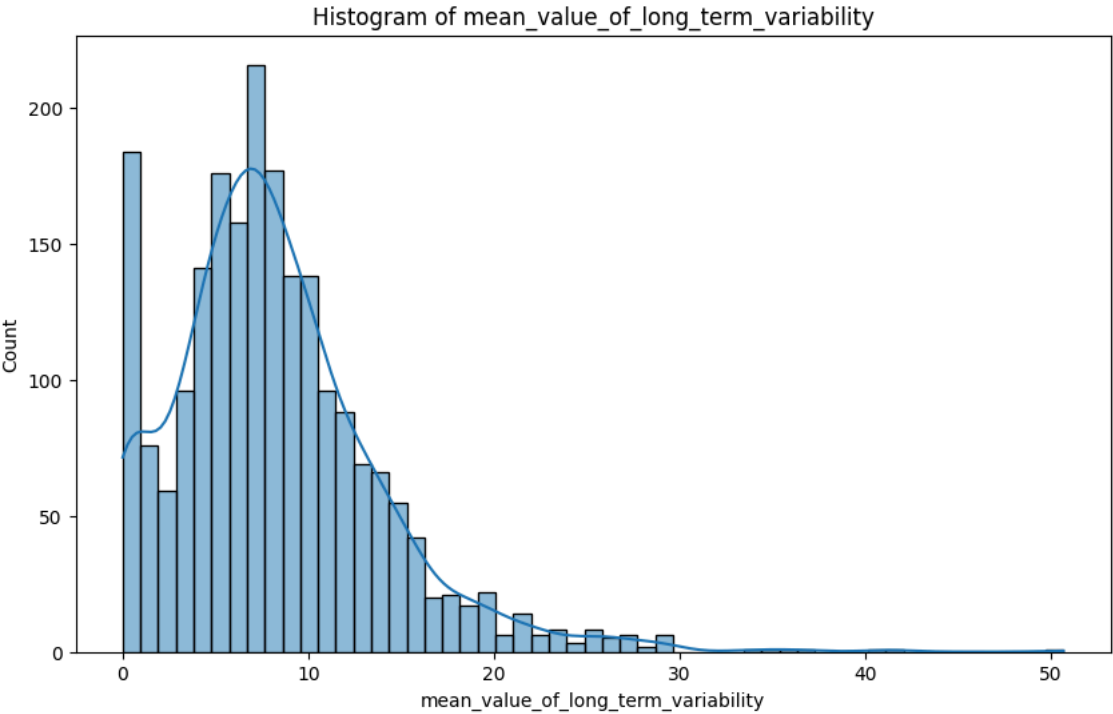


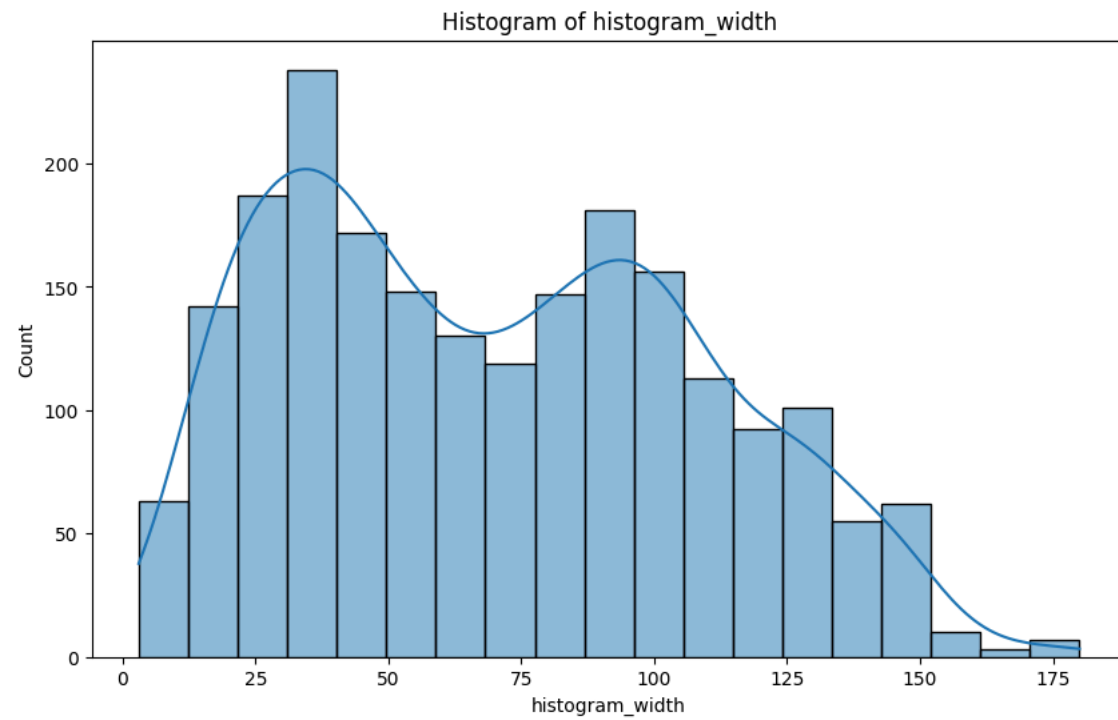


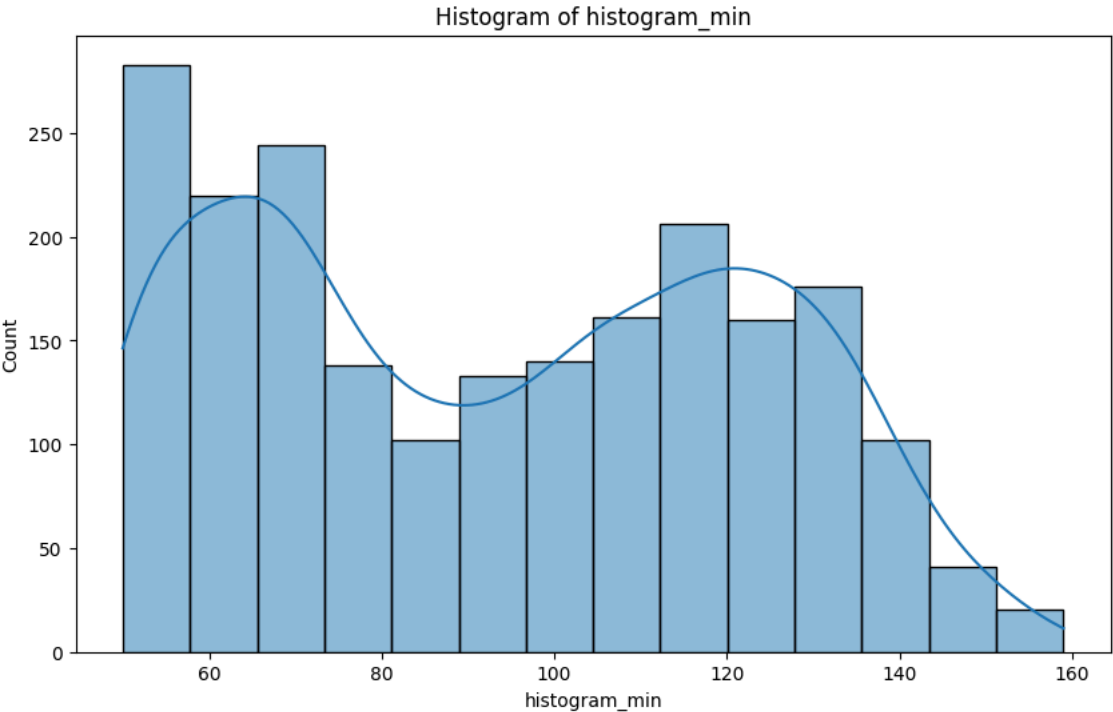


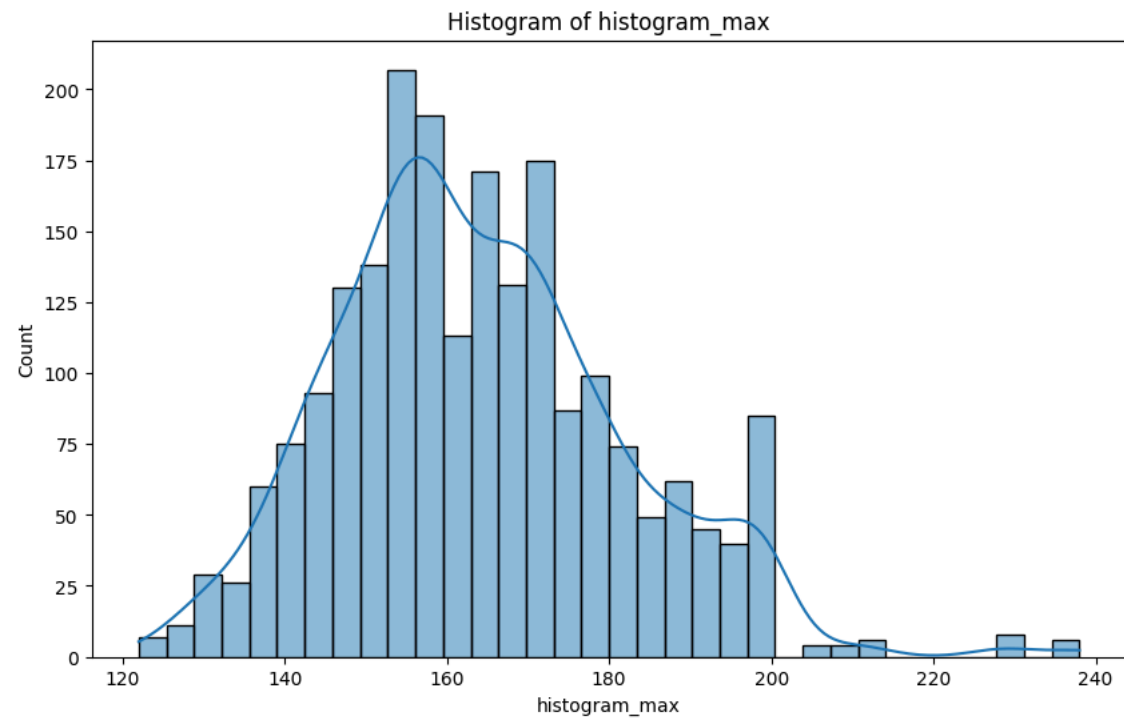


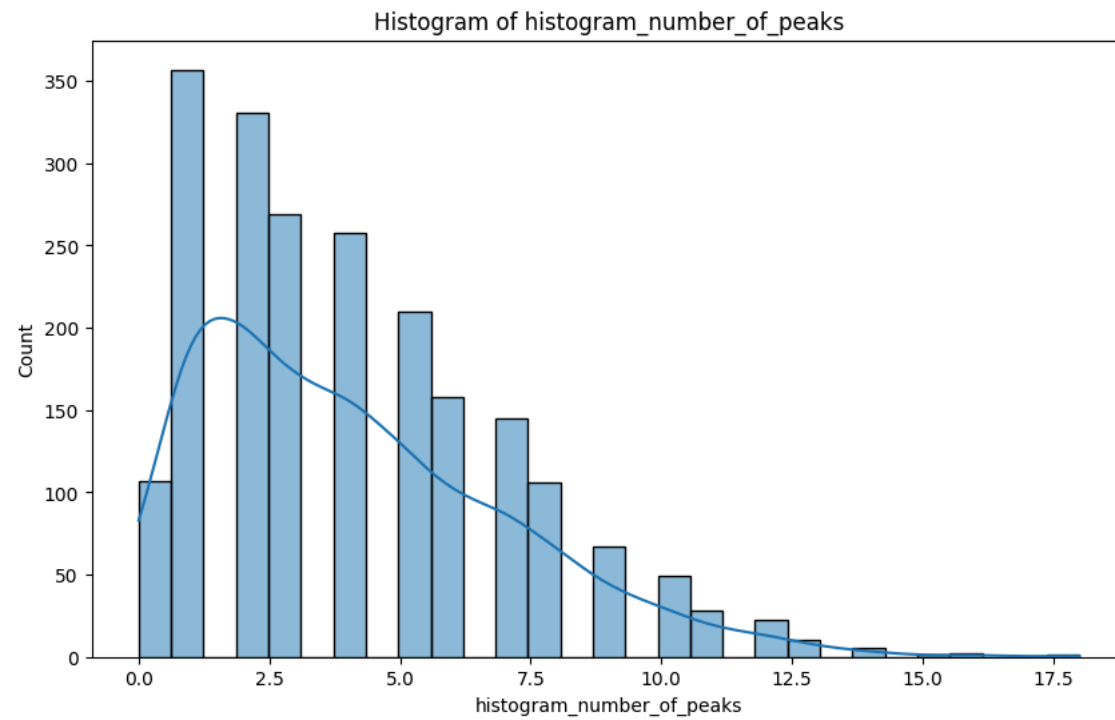


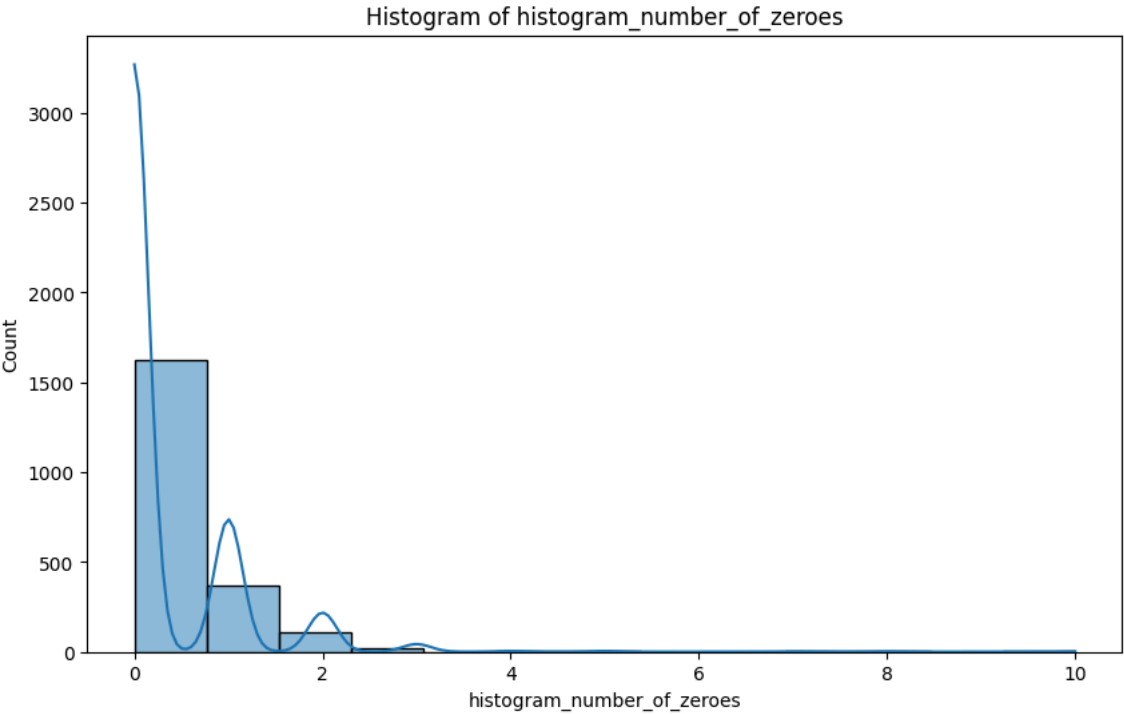


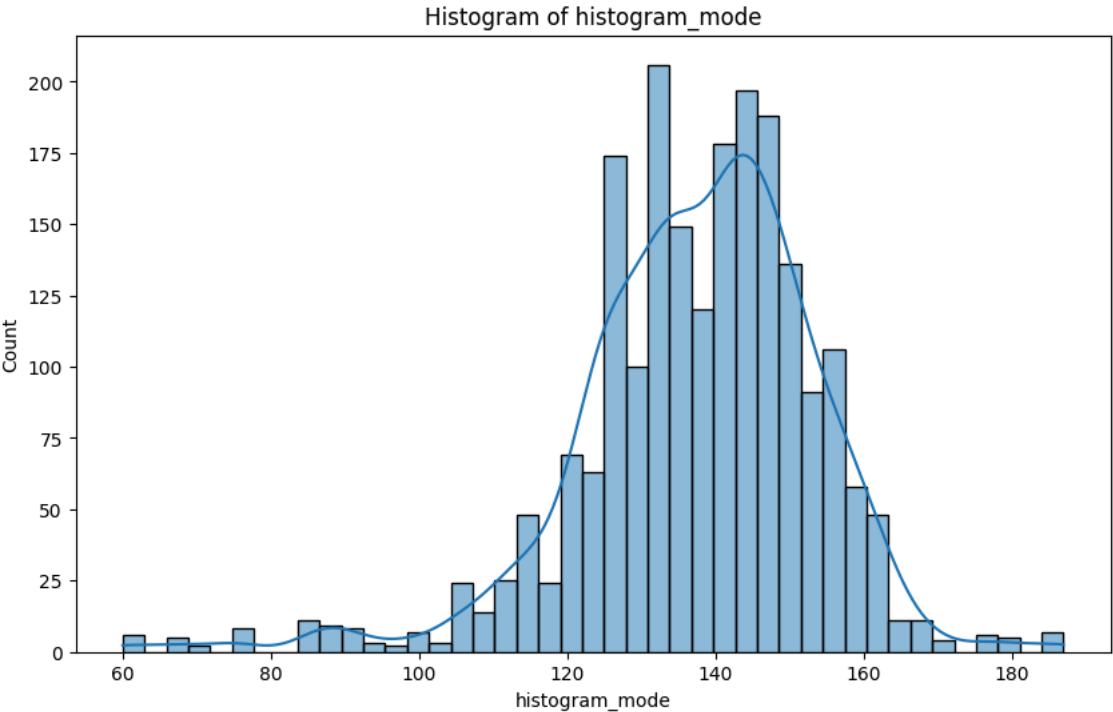


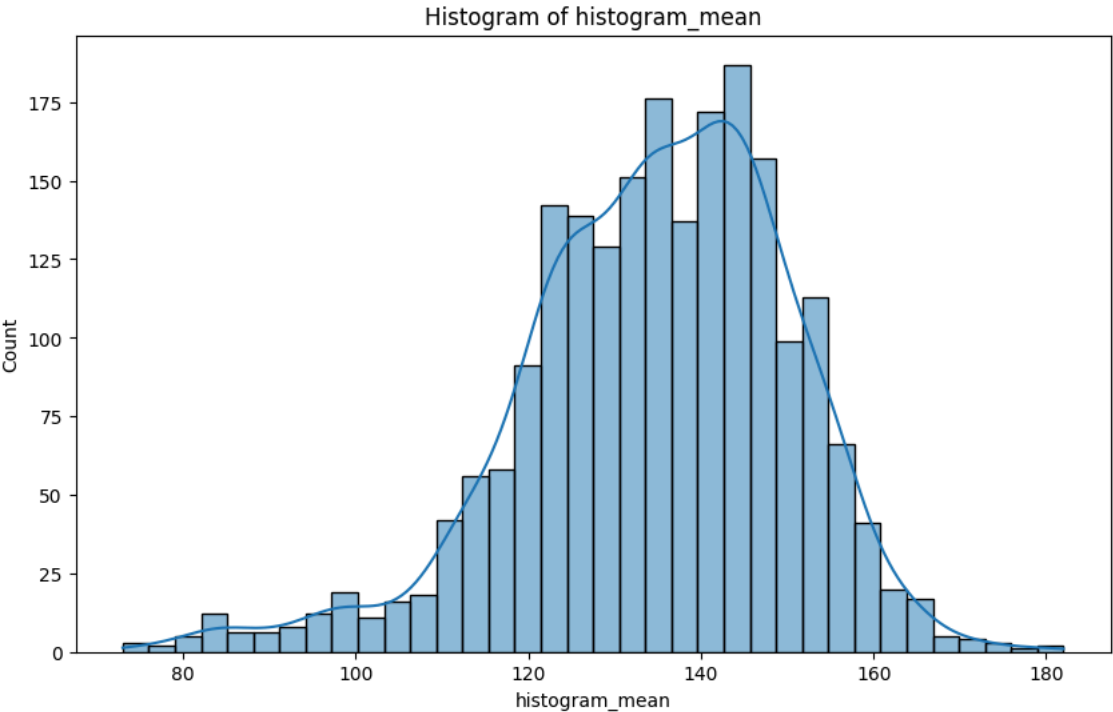


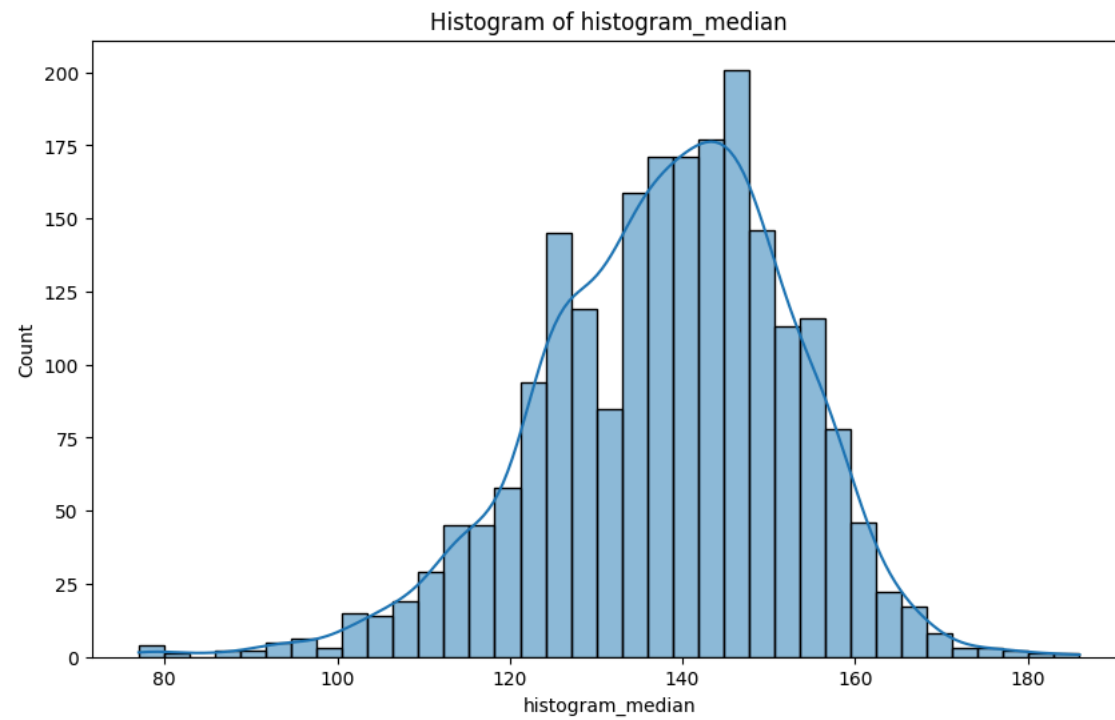


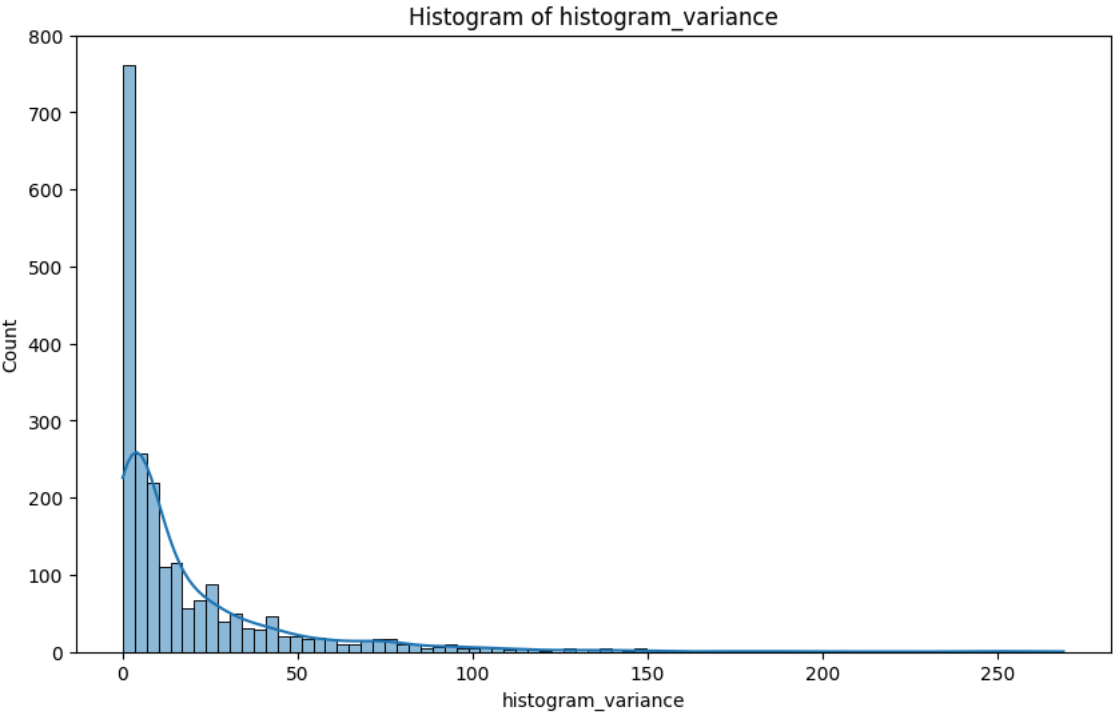


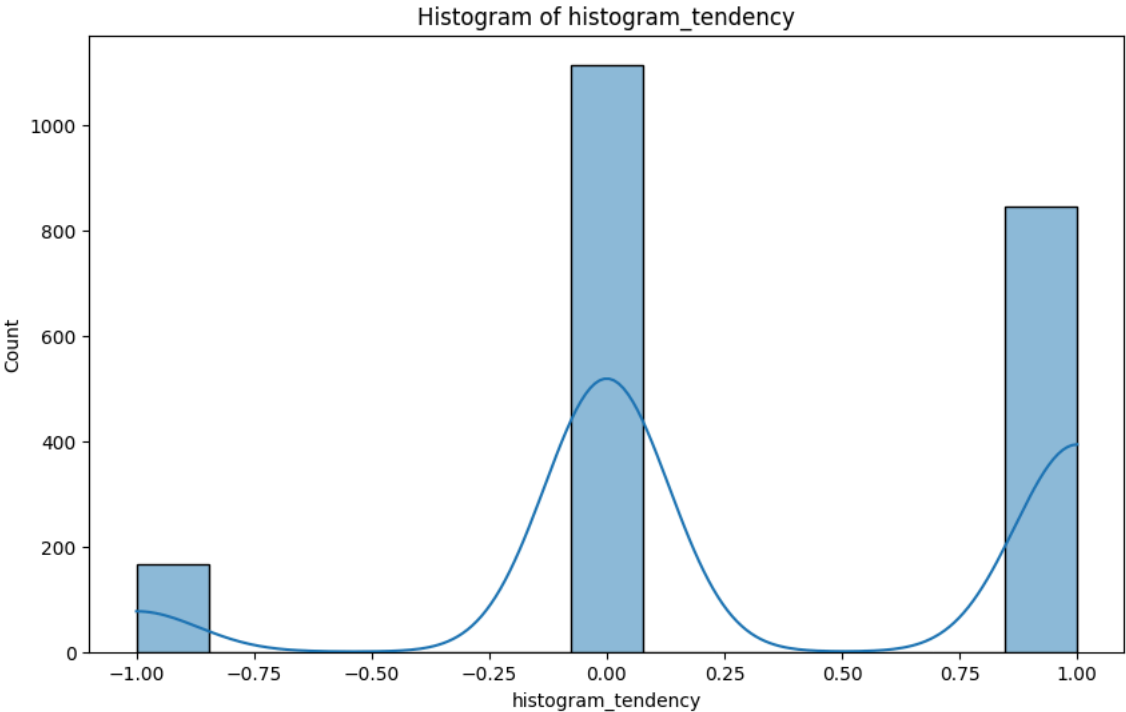


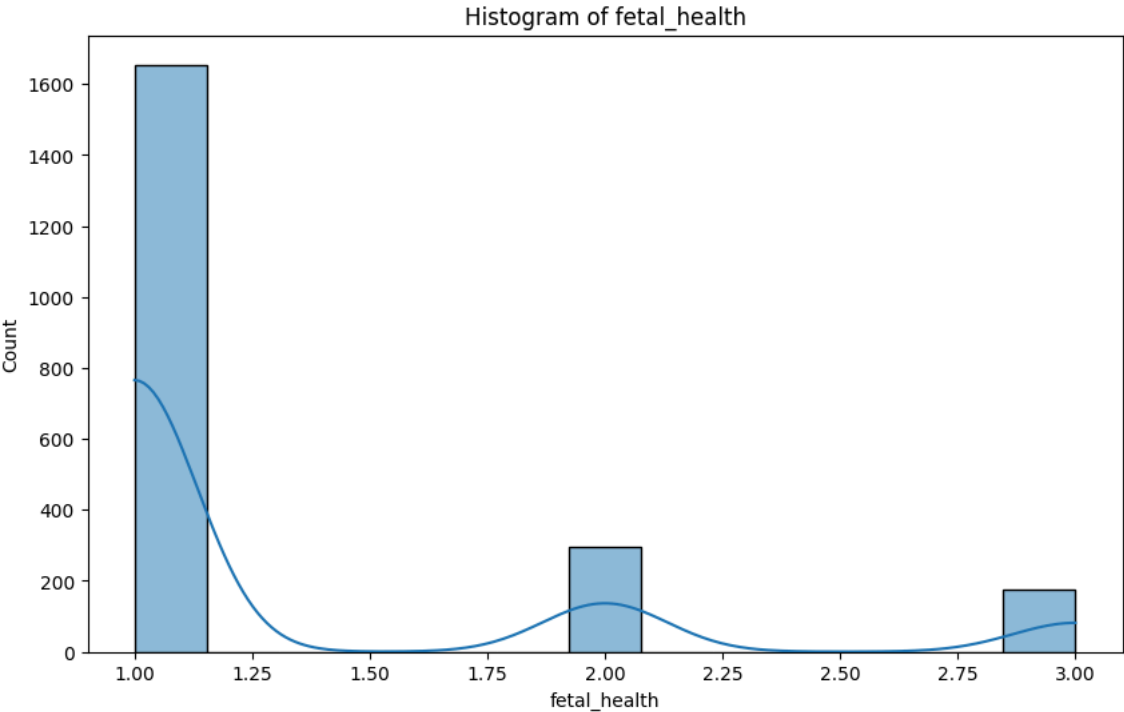




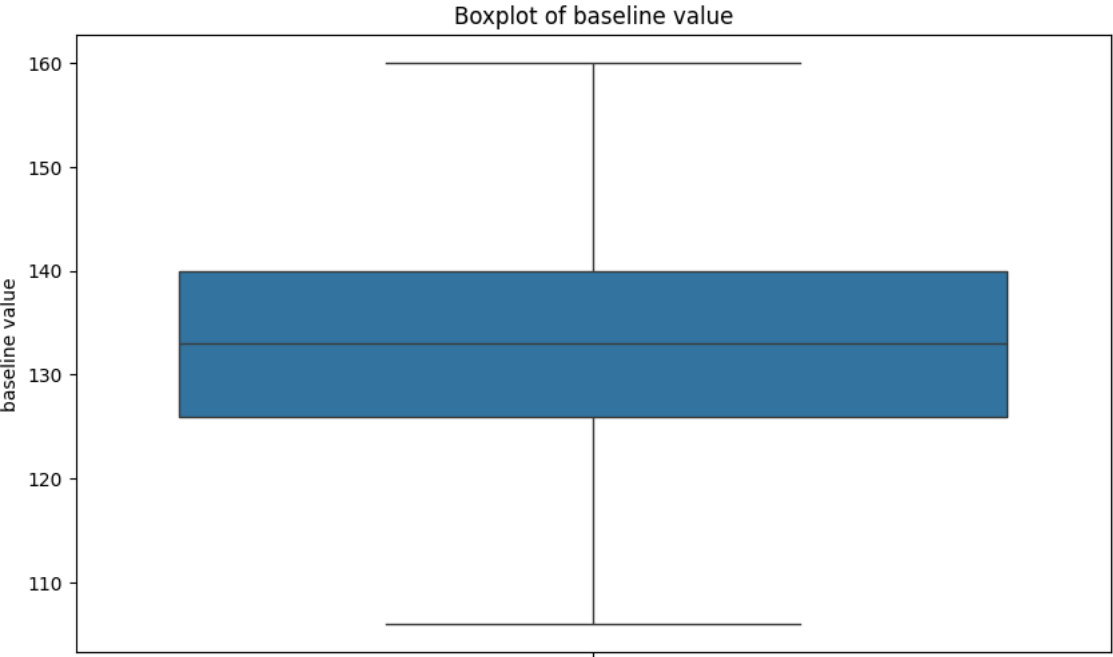




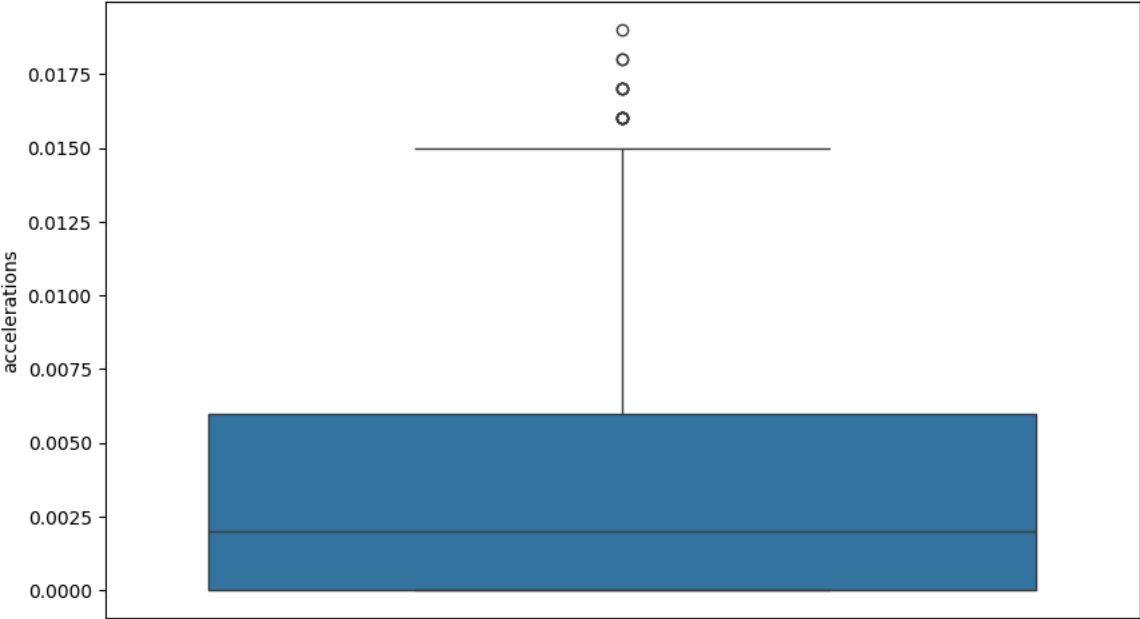




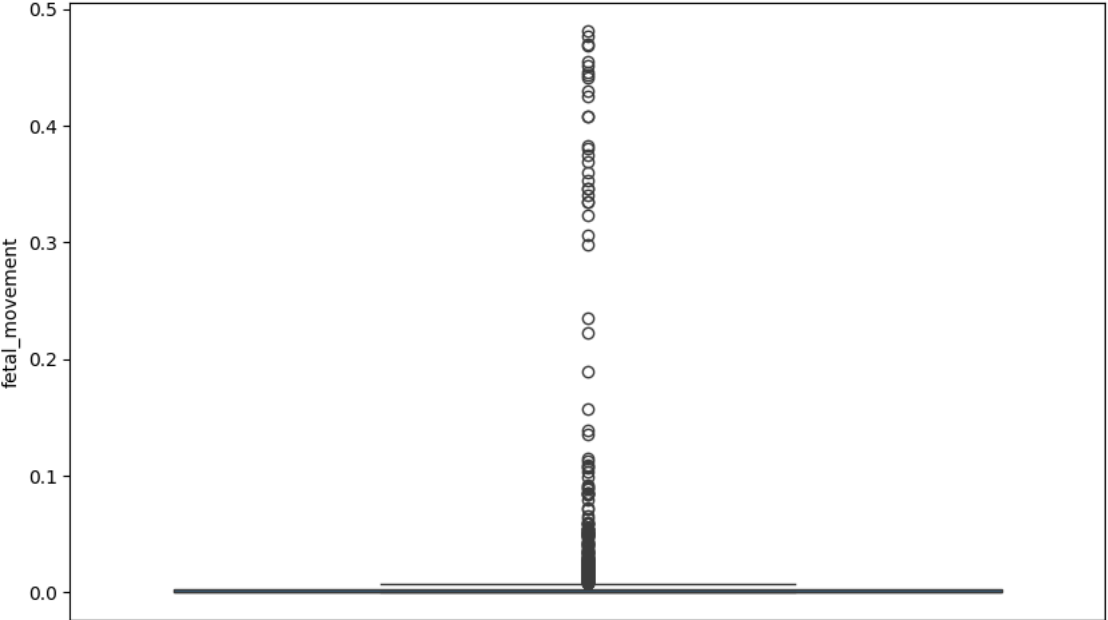
'Boxplots for numerical columns:'



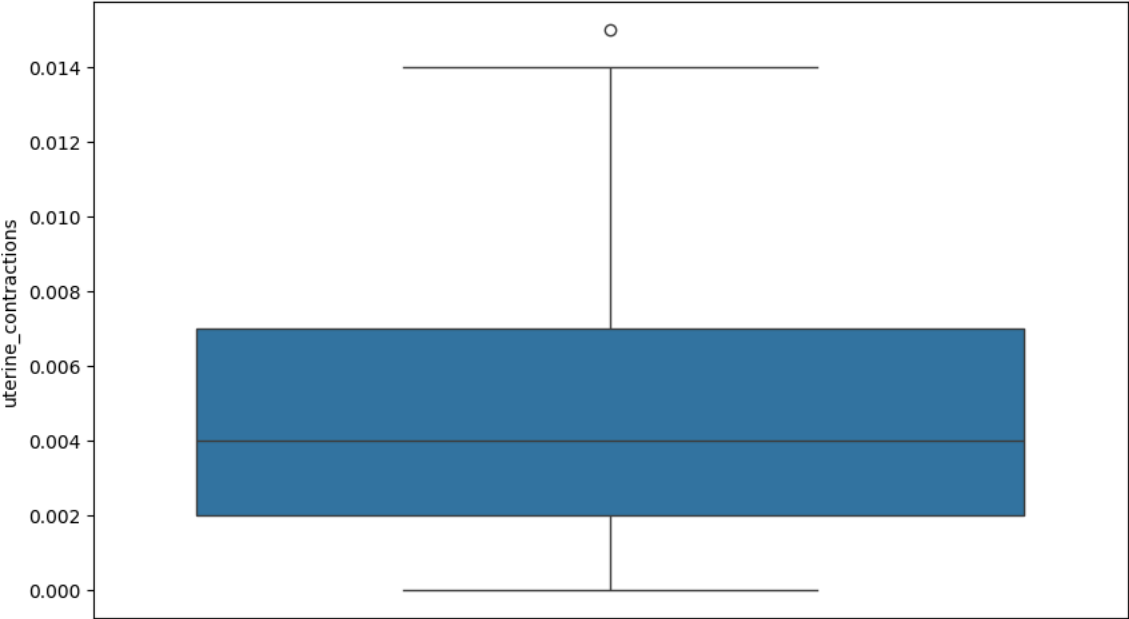
Boxplot of accelerations



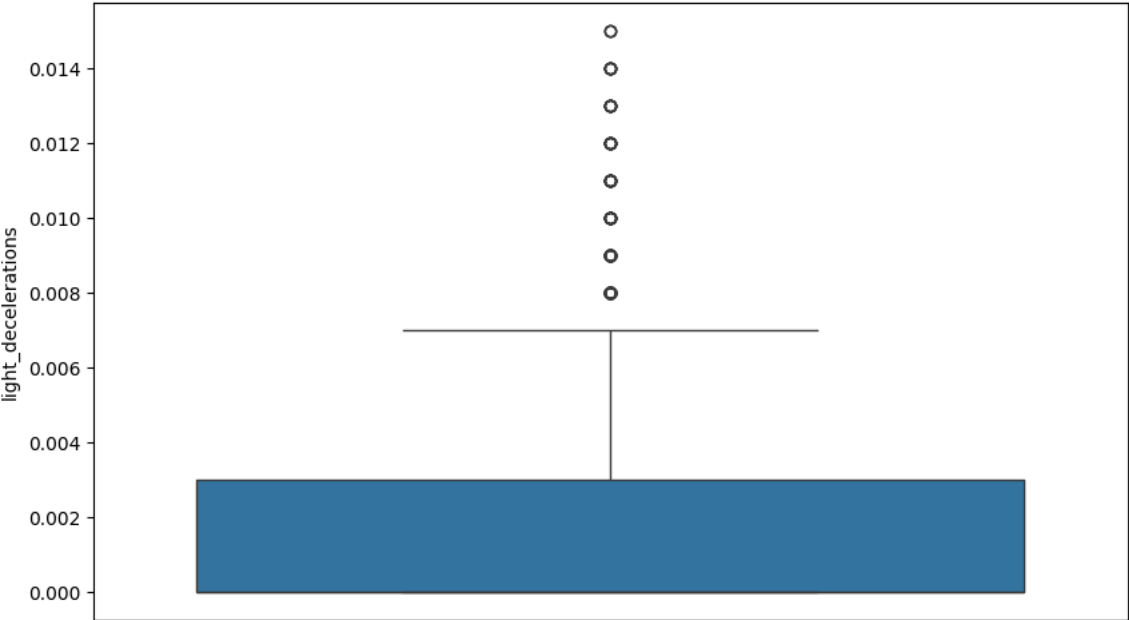
Boxplot of fetal_movement



Boxplot of uterine_contractions

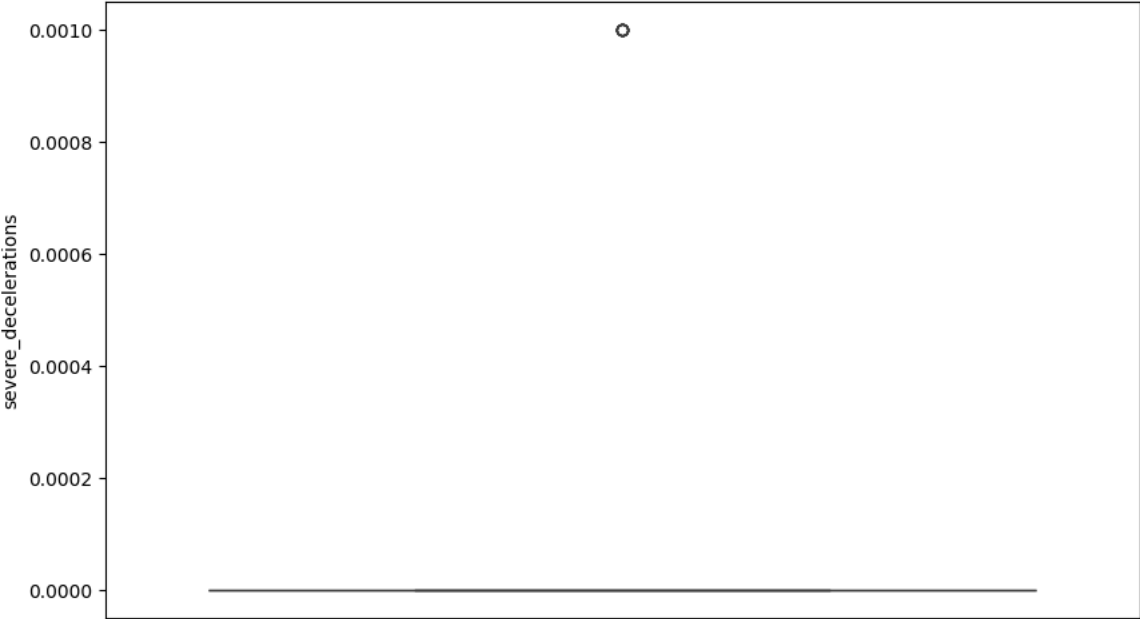


Boxplot of light_decelerations

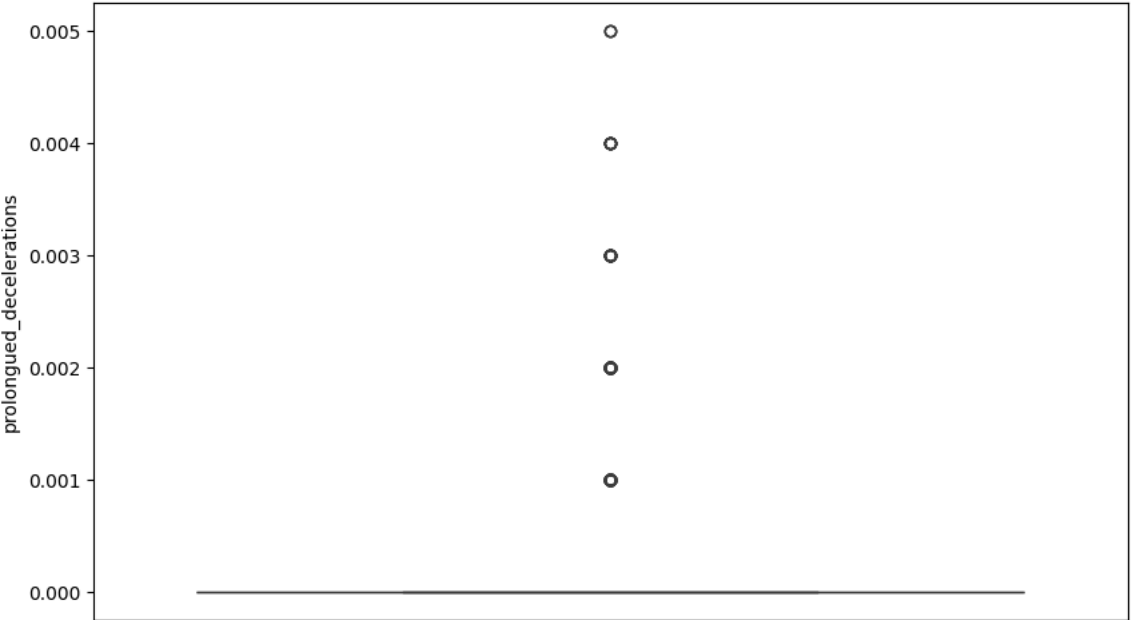


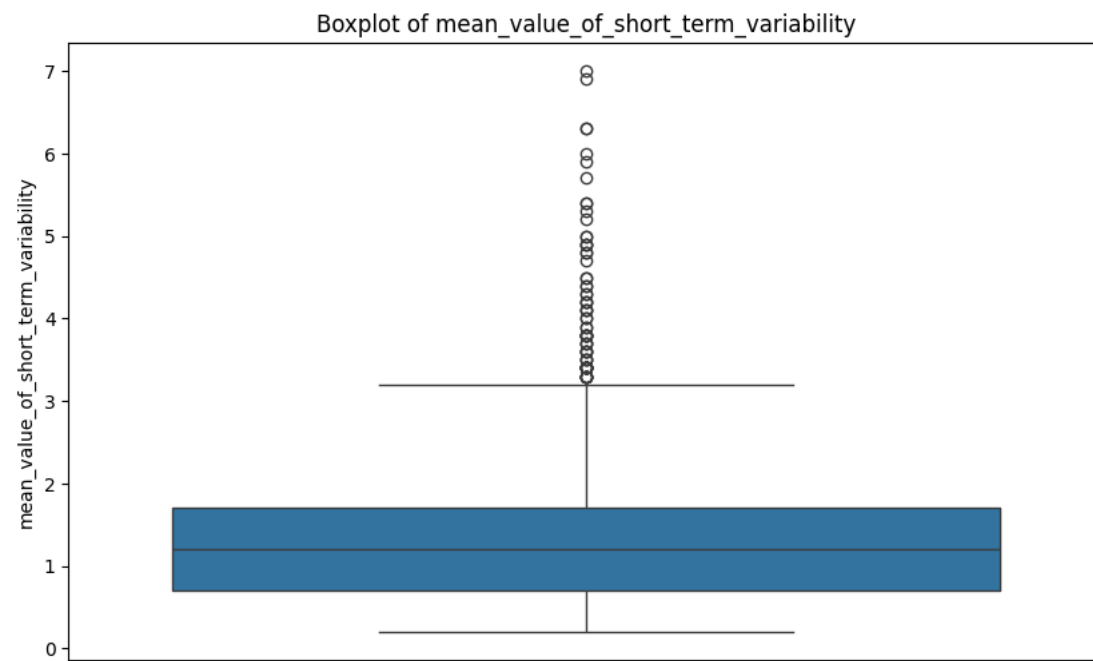
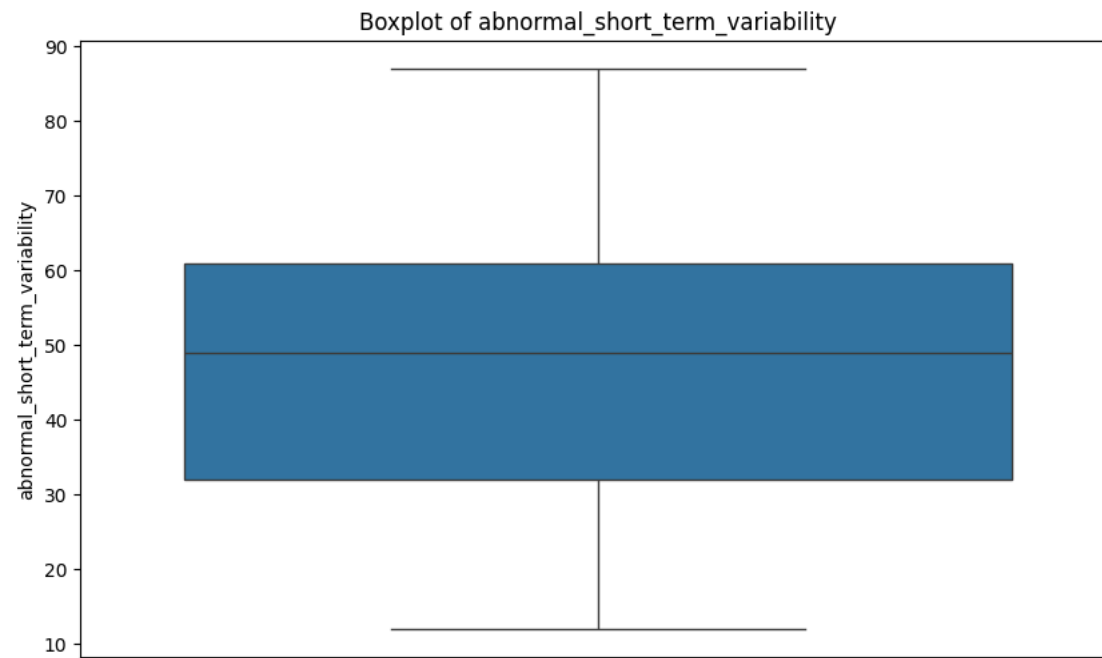


Boxplot of severe_decelerations

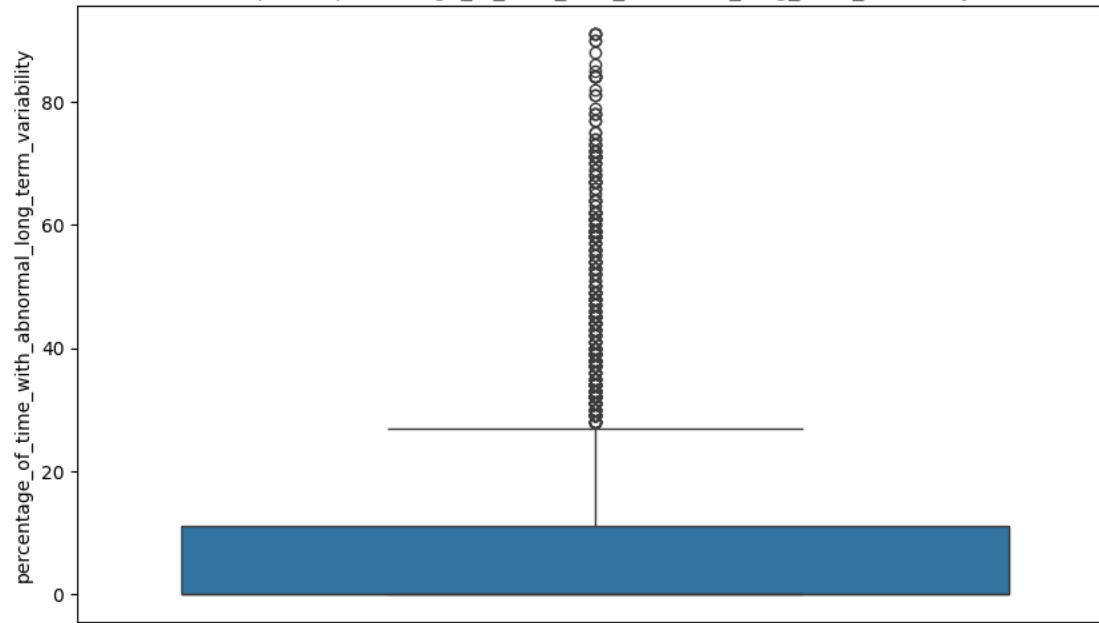


Boxplot of prolonged_decelerations

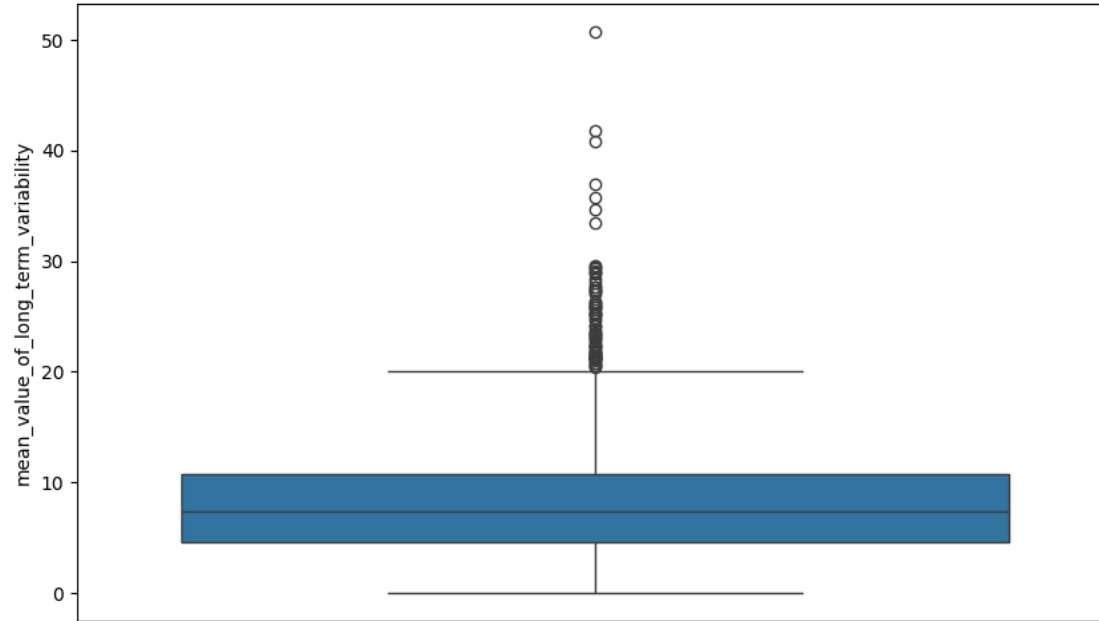




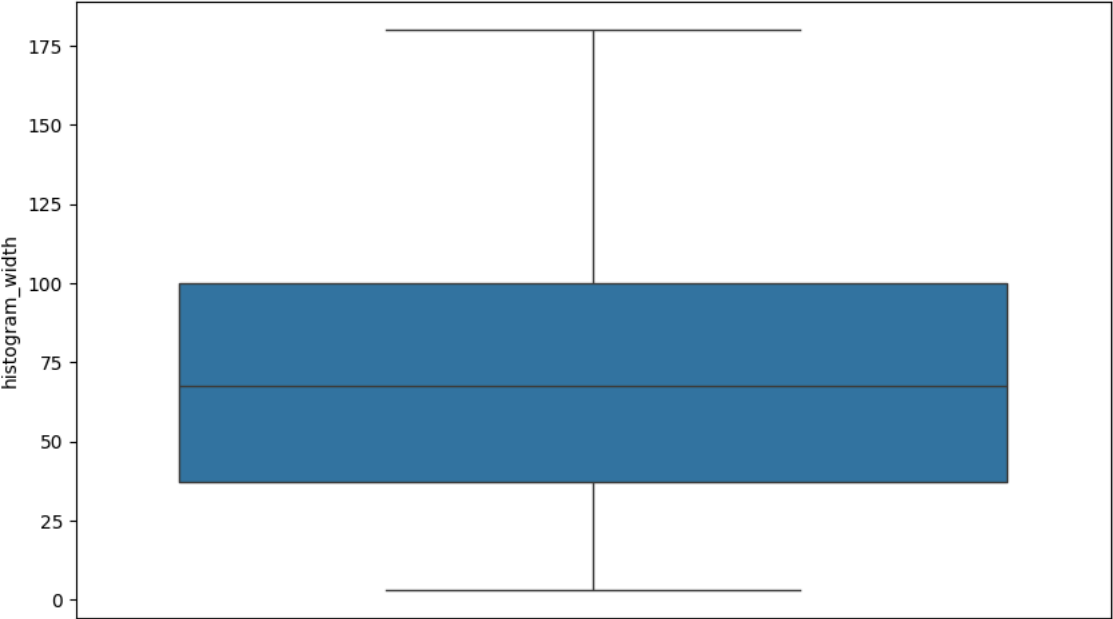
Boxplot of percentage_of_time_with_abnormal_long_term_variability



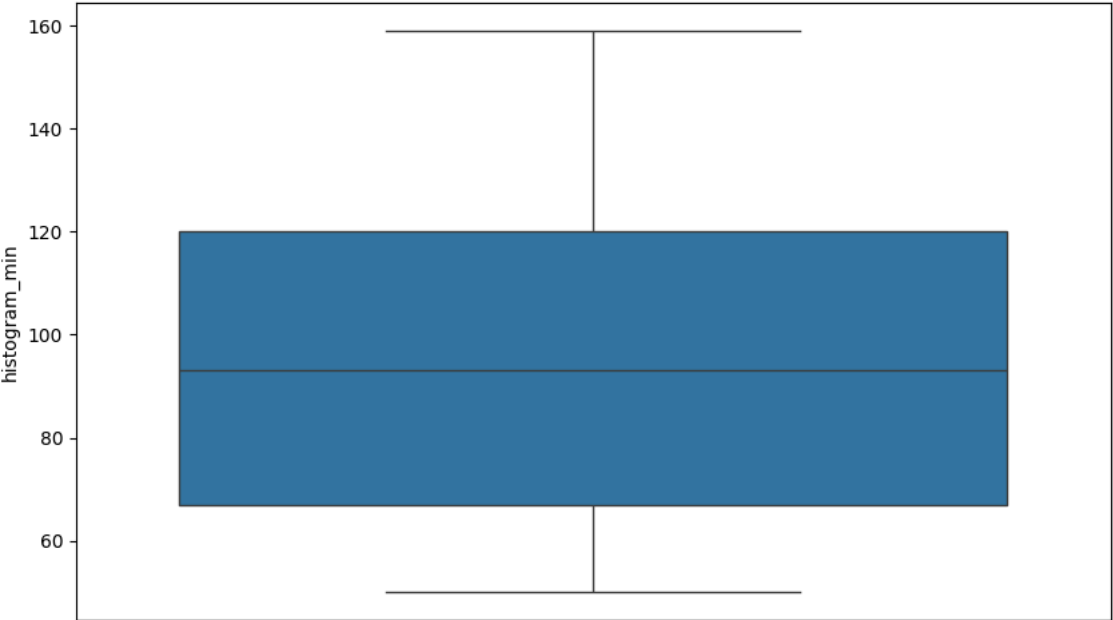
Boxplot of mean_value_of_long_term_variability

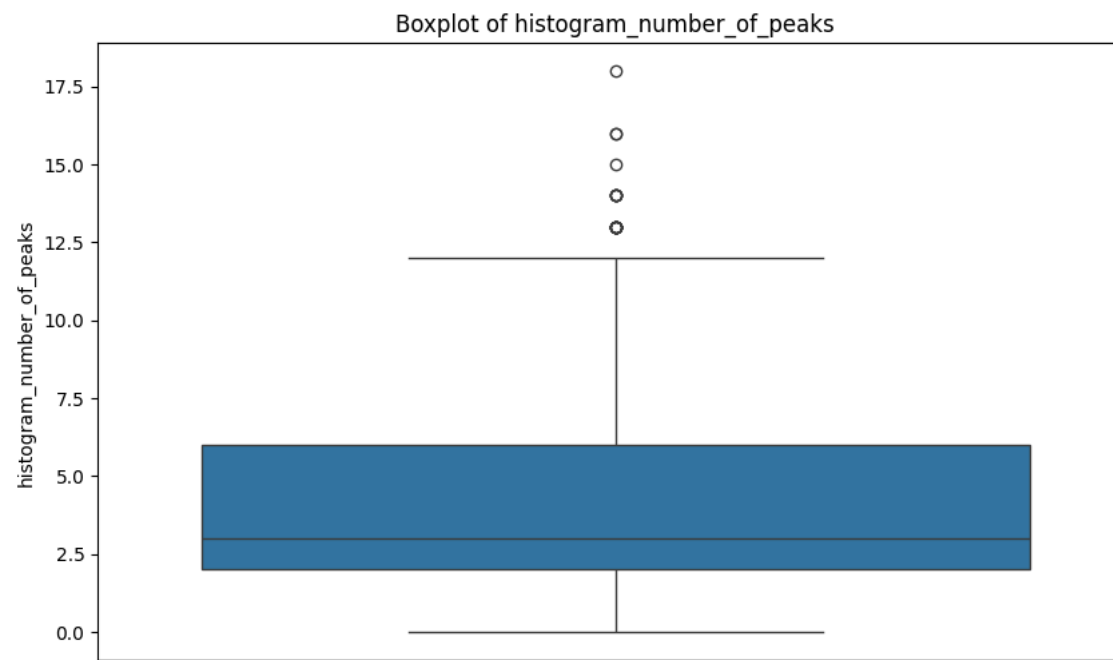
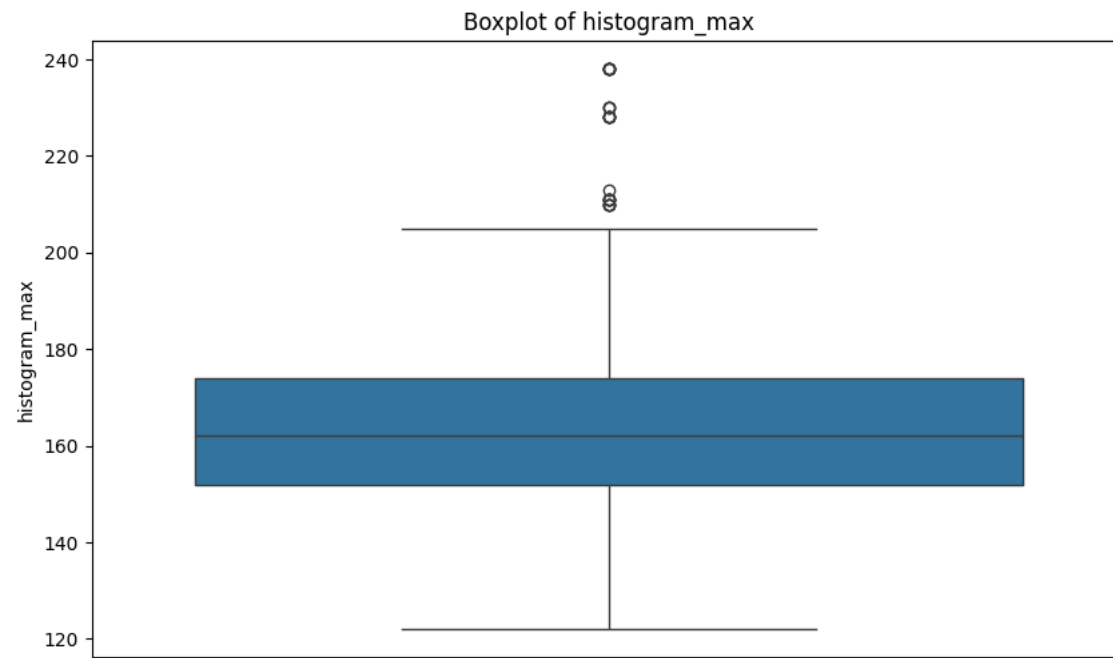


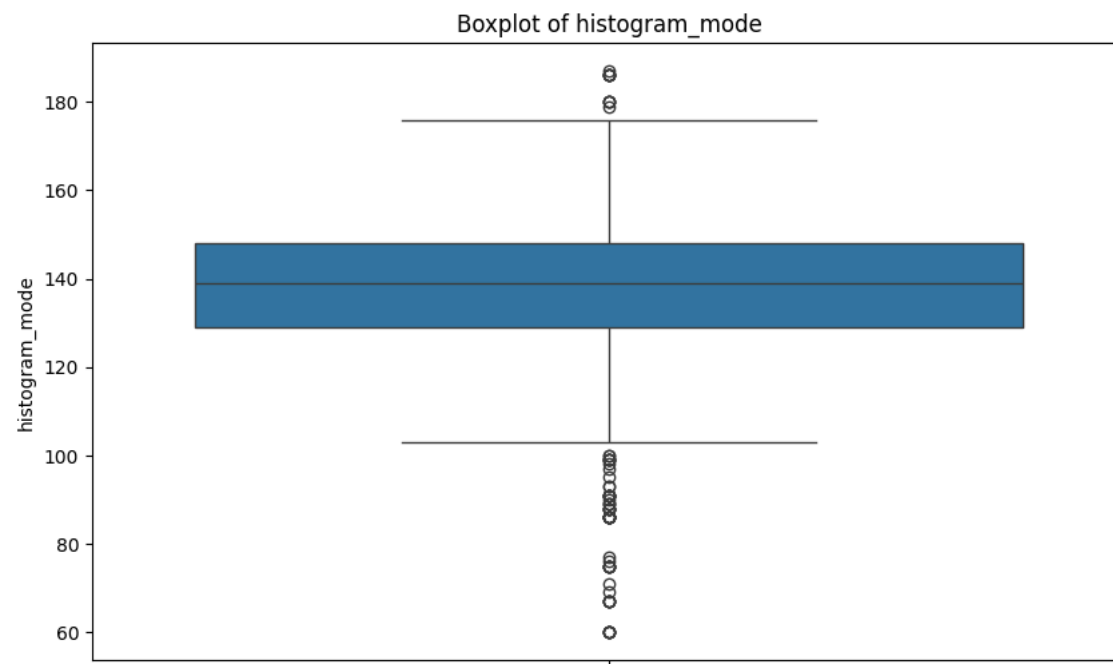
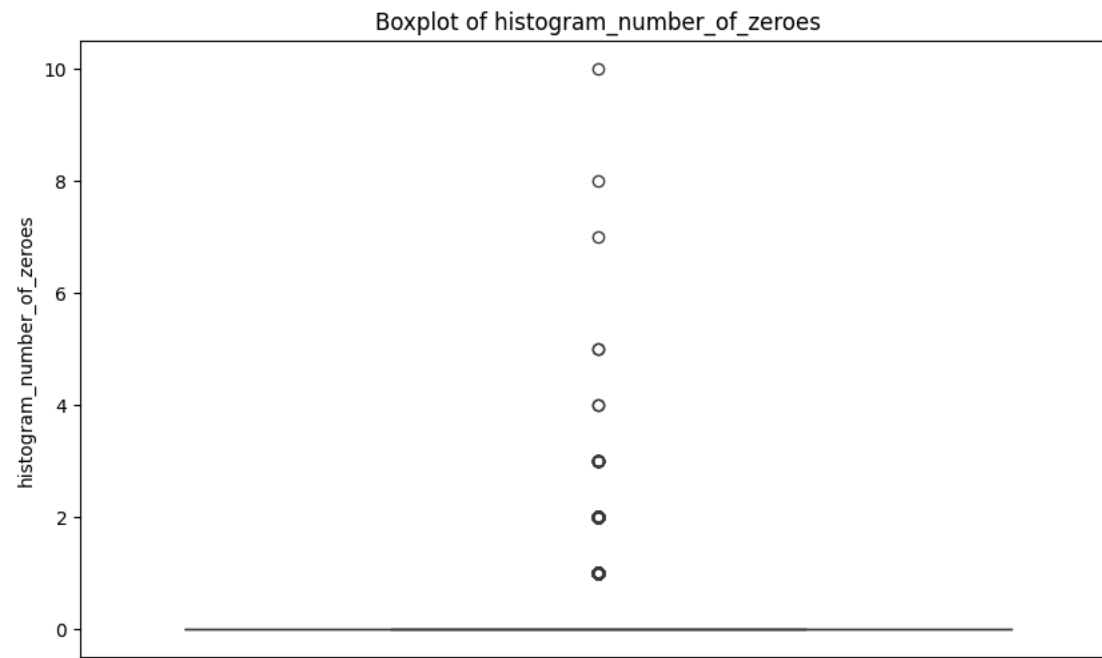
Boxplot of histogram_width



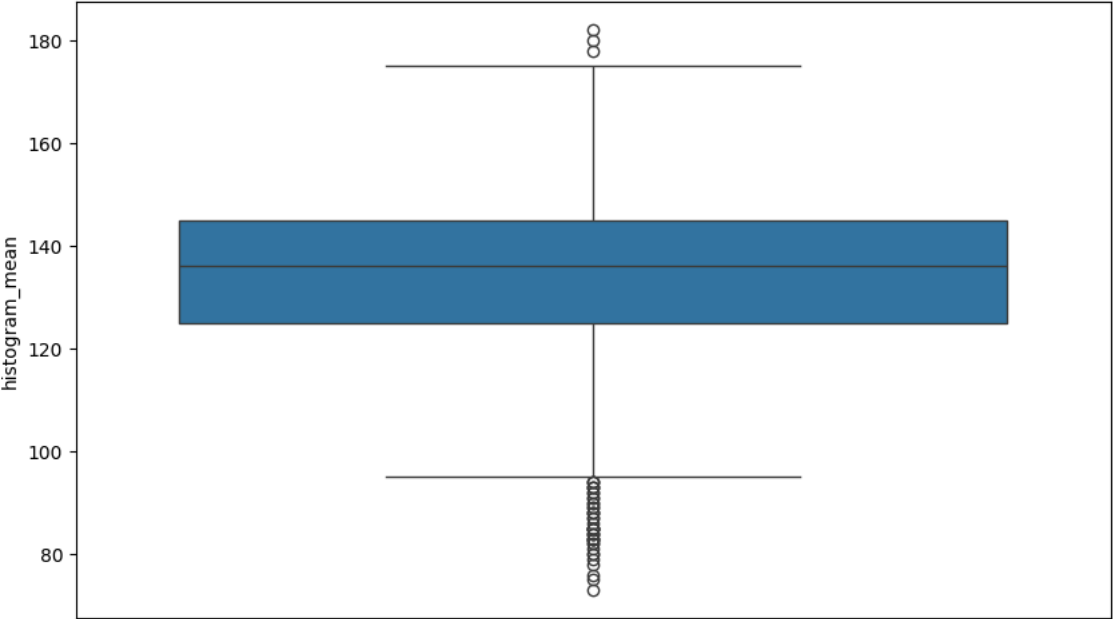
Boxplot of histogram_min



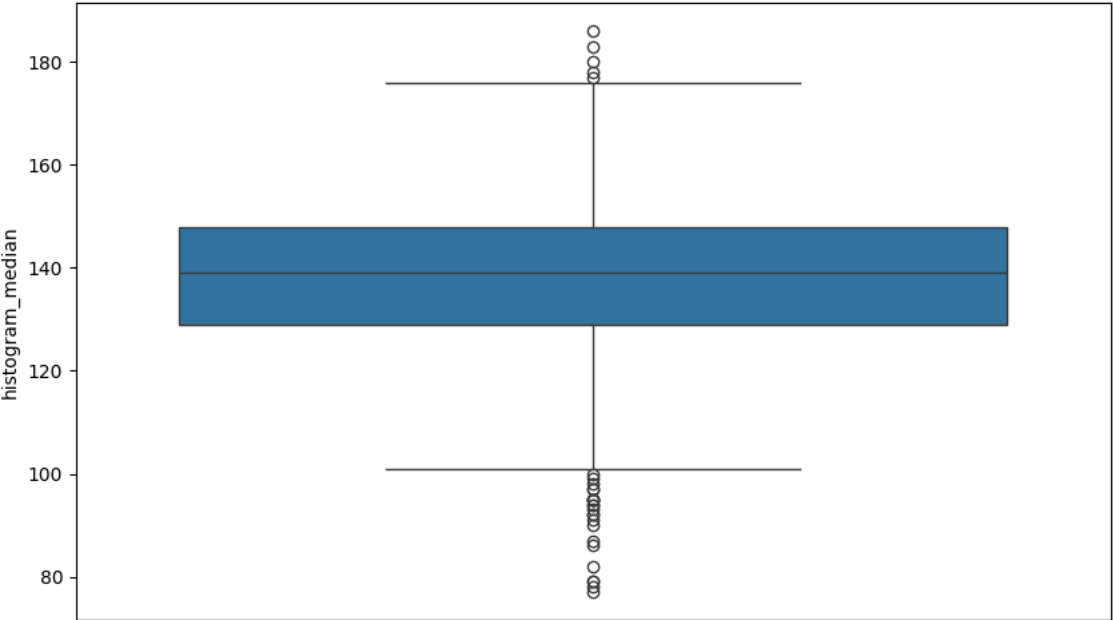


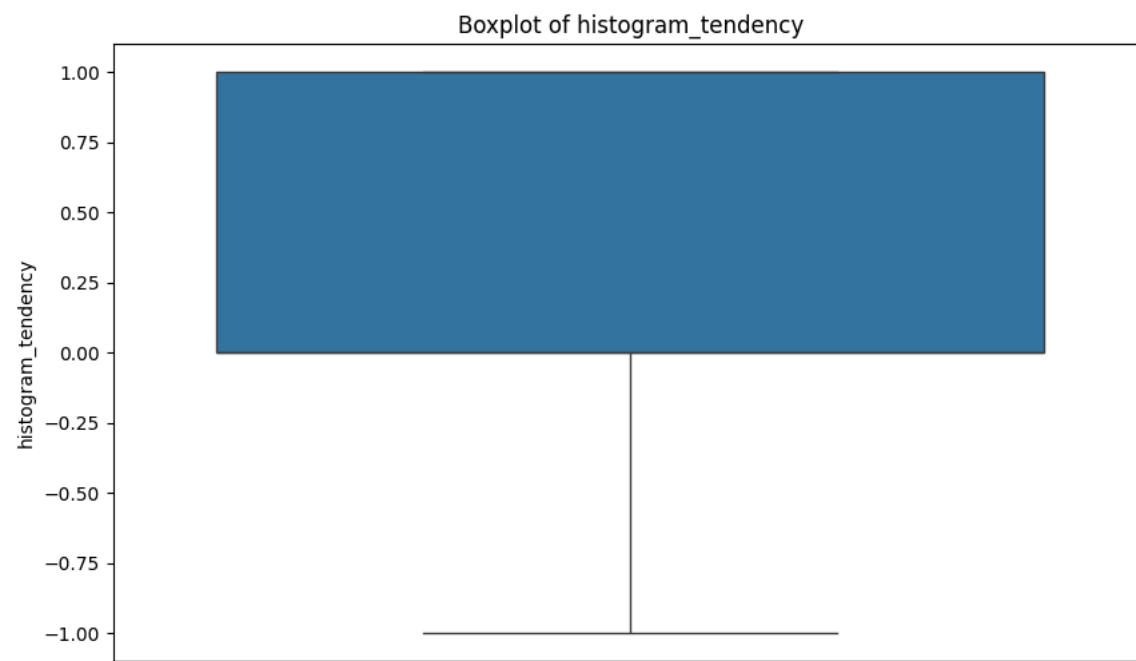
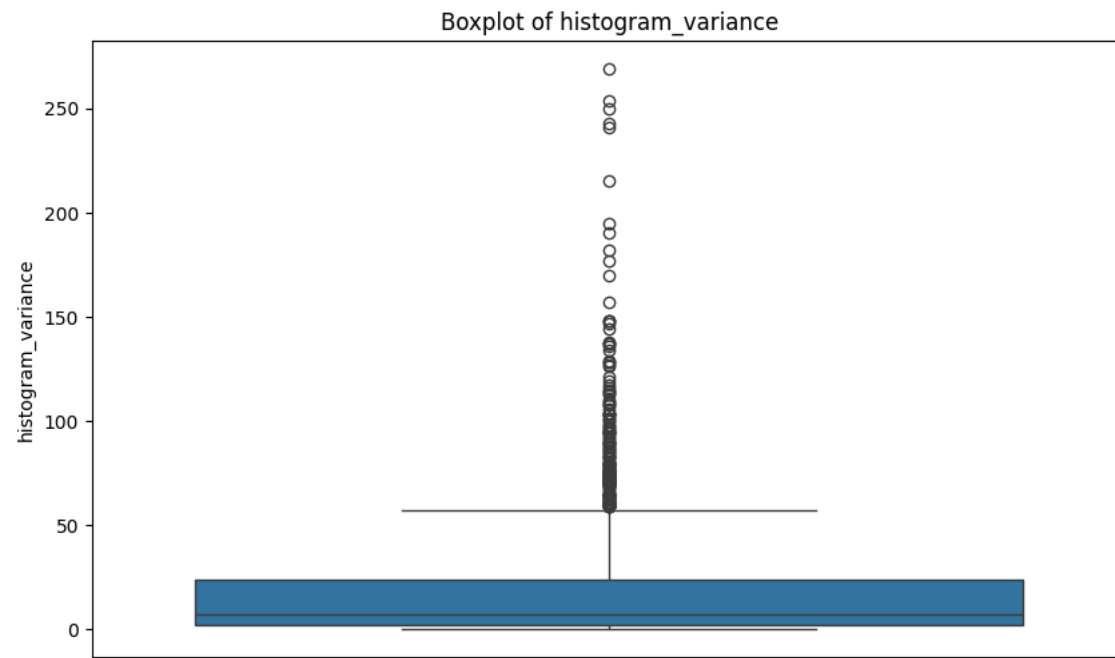


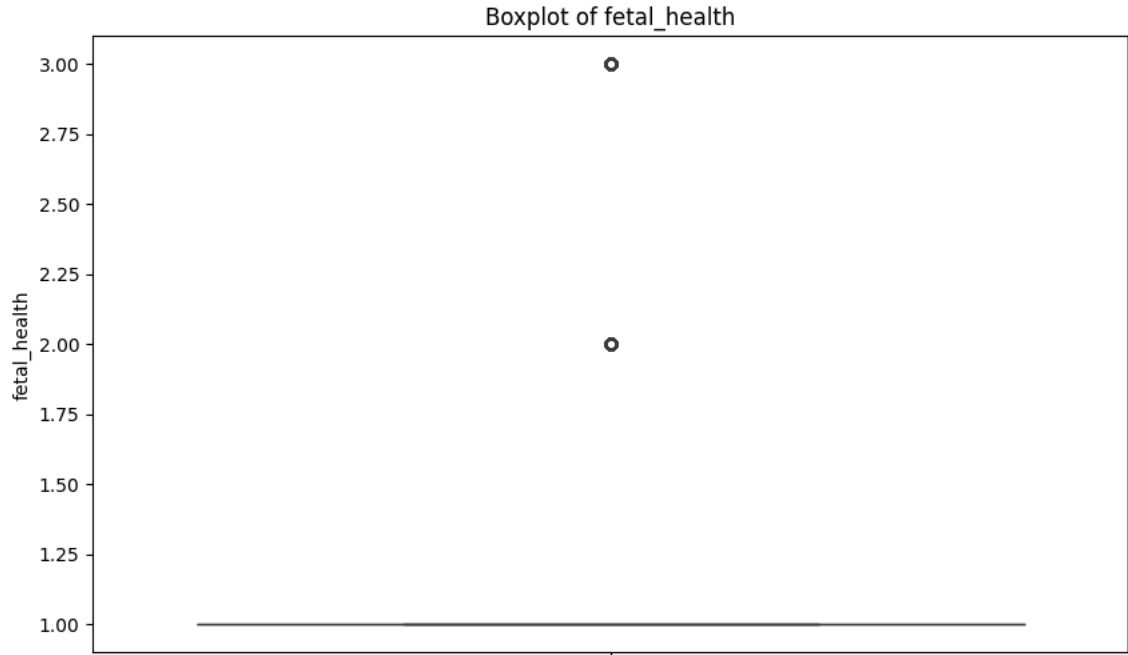
Boxplot of histogram_mean



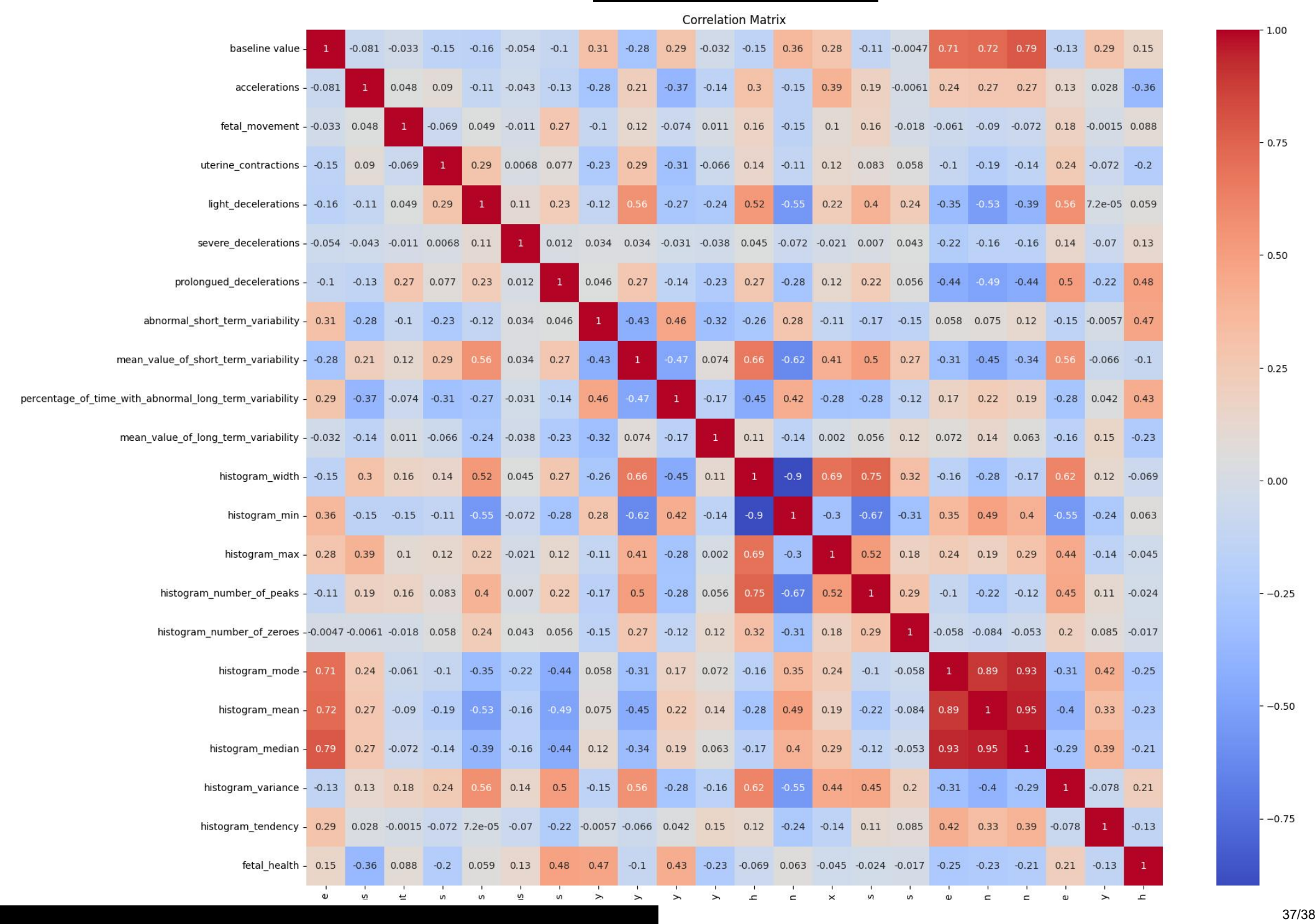
Boxplot of histogram_median







'Correlation matrix: '



'Highly correlated features:'
I tried to run the code, but got the following error: name 'np' is not defined

| | |
|--|--|
| baseline valu | |
| acceleration | |
| fetal_movemer | |
| uterine_contraction | |
| light_deceleration | |
| severe_deceleration | |
| prolongued_deceleration | |
| abnormal_short_term_variablilit | |
| mean_value_of_short_term_variablilit | |
| percentage_of_time_with_abnormal_long_term_variablilit | |
| mean_value_of_long_term_variablilit | |
| histogram_widt | |
| histogram_mi | |
| histogram_ma | |
| histogram_number_of_peak | |
| histogram_number_of_zeroe | |
| histogram_mod | |
| histogram_mea | |
| histogram_media | |
| histogram_varianc | |
| histogram_tendenc | |
| fetal_healt | |